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Photogrammetric acquisition protocol

F.O.P.P.A. (Functional Ontology Protocol for Photogrammetric Acquisition)

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In the context of research for the PhD in Tech4Culture at the University of Turin, 38th Cycle, supervisor Vincenzo Lombardo

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Overview

Functional Ontology Protocol for Photogrammetric Acquisition (F.O.P.P.A.) proposes a structured approach to the management of photogrammetric data in archaeological contexts. Recognizing the multidisciplinary nature of photogrammetry, especially in the field of cultural heritage, F.O.P.P.A. aims to provide a unified methodological framework based on ontological principles. The protocol aims to solve the problems related to the heterogeneity of methods and protocols used in photogrammetric projects, which often hinder communication and interoperability. In particular, it addresses the organizational difficulties caused by the variety and quantity of RAW data collected. A key aspect of F.O.P.P.A. is the integration of CIDOC-CRM, a consolidated ontology in the cultural heritage sector, which provides a shared language for the description of metadata and standardizes the processes related to the management of digital data. This integration ensures continuity between the generative and archiving processes, improving interoperability between projects. F.O.P.P.A. intends to respond to the challenges related to the management of photogrammetric data in the archaeological context, offering a standardized protocol that facilitates traceability, interoperability and effective communication between projects. The objective is to improve the management of photogrammetric data for the conservation of cultural heritage, contributing to the longevity and accessibility of archaeological information.

Purpose of the Manual

1. **Instructions on acquisition processes:** The F.O.P.P.A. manual offers a user-friendly expedition through photogrammetric surveying. Step-by-step guidance, clear language, and visual aids ensure a seamless journey from acquisition to 3D model export.

2. **Exposition of the theoretical aspects of the protocol:**Theoretical elements in the F.O.P.P.A. manual and schematic and image suggestions are strategically placed at the page bottom. This layout ensures seamless integration, letting users align theory with practical implementation at each step for comprehensive understanding.

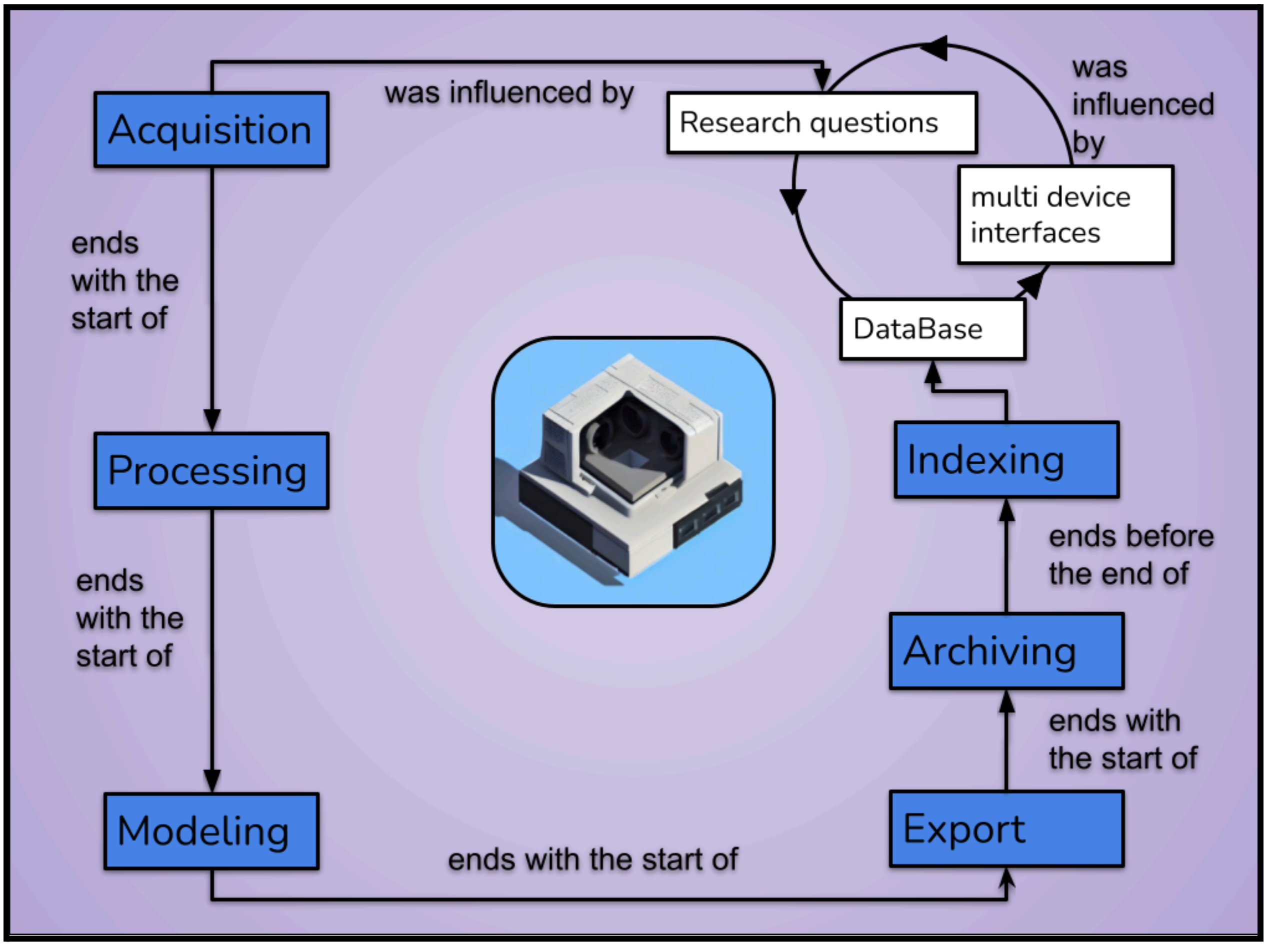
3. **Reference publications:** An exhaustive list of all publications taken into consideration in the creation of the protocol.

Minimum equipment requirements

F.O.P.P.A. is designed to be accessible and adaptable, requiring only a modest set of equipment to embark on photogrammetric endeavors. The primary tool is a camera, and even compact models boasting a minimum of 16 megapixels can yield satisfactory results. This inclusivity allows practitioners with varying budgets and equipment access to the benefits of the protocol. Additionally, a computer equipped with at least an i5 processor or an equivalent is recommended to handle the computational demands of photogrammetric processing. Notably, the protocol is agnostic regarding processing and modeling software, recognizing the diversity of user preferences and project needs. While software recommendations include Agisoft, Autodesk, Zephyr, Regard 3D for processing, and Blender, Maya, or equivalent for modeling, F.O.P.P.A. remains versatile, accommodating various software choices based on user familiarity and project requirements. It is crucial, however, for users to possess a foundational understanding of photogrammetry and modeling software, as the subsequent sections of the F.O.P.P.A. manual delve into specific operations, assuming a basic proficiency in these tools. This flexibility ensures that F.O.P.P.A. remains a widely applicable protocol for diverse practitioners engaged in cultural heritage preservation and archaeological documentation.

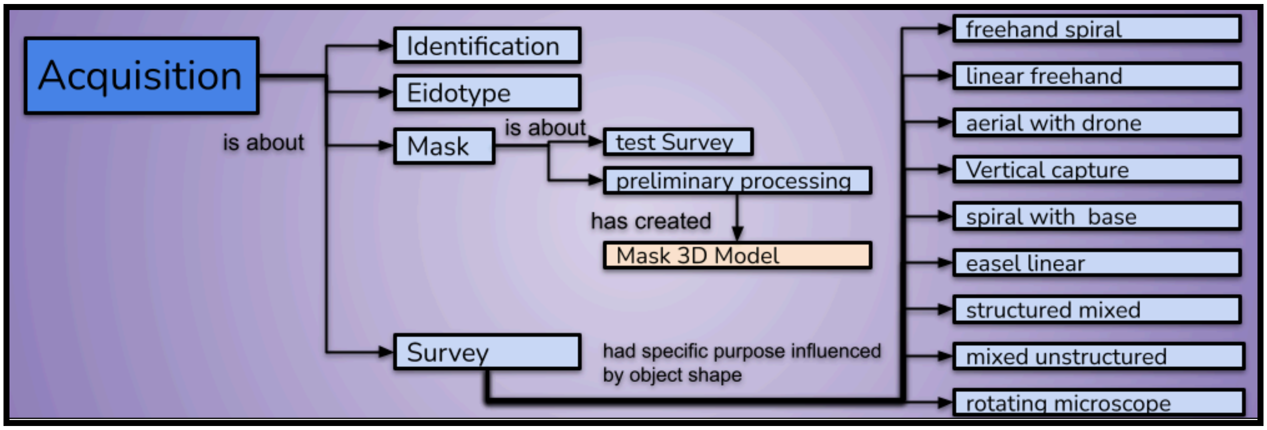
General diagram of the acquisition process

The Acquisition phase in the F.O.P.P.A. protocol represents the foundation of the entire photogrammetric process. The focus is on the photographic capture of archaeological objects, with a structured approach that includes activities such as identification, drawing of the Eidotype, creation of the mask and photogrammetric survey. Identification focuses on the recognition of the main geometric shapes of the object, while the drawing of the Eidotype provides a visual reference map. The creation of the mask frames the area to be surveyed, ensuring complete data coverage, and the photogrammetric survey is carried out following a movement plan based on the geometry of the object, respecting the "rule of opposites" to prevent geometric distortions. Subsequently, the protocol enters the Processing phase, where the validation of the acquired data takes on a central role. The focus is on verifying the homogeneity and quality of the data, in line with the chosen algorithms, without imposing the use of specific software. The Processing phase provides a structured approach to address any issues, while maintaining the integrity of the acquired data. The Modeling and Export phases mark the conclusion of the modifications to the 3D model and the beginning of its final definition. Although the use of specific software for modeling or export is not imposed, the protocol emphasizes the importance of maintaining consistency with the previously selected algorithms and with the parameters of storage and final use. This phase ensures a smooth transition of the 3D model to its final destination, which may include a database, a virtual museum or a research context for further analysis. Data Management is a crucial aspect of the F.O.P.P.A. protocol, which addresses the management of archives generated by photogrammetric processes. The protocol recommends an organized approach based on CIDOC-CRM, which ensures standardized storage methods and facilitates future accessibility, verification and interoperability between different projects. Finally, the Indexing phase focuses on the systematic organization and cataloging of archived data. F.O.P.P.A. introduces a methodical structure for indexing, where each 3D model is associated with a dedicated folder structure, including subfolders for photos, processing projects and modelling projects. The nomenclature assigned to each model reflects relative chronologies, archaeological typologies, individual identifiers and contextual variants. This approach ensures the orderly availability of RAW data and their consistency with acquisition methods and archived information.



Acquisition

The F.O.P.P.A. protocol's Acquisition Phase initiates the digital documentation journey with the Identification Sub-Phase, involving a thorough study of the archaeological object. This includes recording catalog numbers, excavation details, and material characteristics, alongside an evaluation of environmental factors. Subsequently, the Mask Relief Sub-Phase assesses the camera's behavior towards light sources, aiding in studying illumination within a set. The Eidotype Drawing Sub-Phase introduces a logical and graphic process, outlining the general shape and critical points of the object. This step facilitates determining the acquisition type based on the object under study. The core of the Acquisition Phase is the Survey Sub-Phase, where the actual survey takes place. The number of photographs depends on hardware and software capabilities, with considerations for light positions to preserve color information. The Acquisition Phase concludes with various techniques like Straight Line, Spiral, Aerial, and Vertical, tailored to specific scenarios for a comprehensive digital representation. The meticulous approach in this phase ensures fidelity to the original archaeological context, setting the stage for subsequent stages in the F.O.P.P.A. protocol.



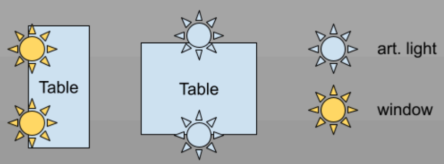
Identification

RUNNING EXAMPLE

Identification

Identification:
Collection of

- the Archaeological Name: Yaioi Boar Jaw Min_OkaAr_Fra_Yayoi_BeA005
- the conservation position
  - name: 岡山市埋蔵文化財センター Okayama City Buried Cultural Properties Center
  - UTM coordinates: 34.64580275826324, 133.93813493465072) annotation of the reference culture: Yayoi
- the absolute dating offered by the archaeologists: 200 P.E. ca.
- Study of the position for the survey in reference to windows and lighting and bone material.



- Study of the museum's opening hours and availability of time for the survey: in this case 35 minutes
- Annotation for the data of the Survey: 09/03/2019

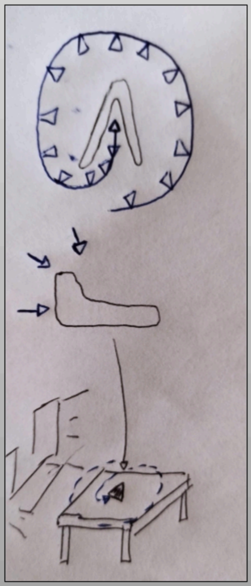
The commencement of the acquisition phase heralds the crucial Identification sub-phase in the F.O.P.P.A. protocol, a meticulous process ensuring a methodical start to the photogrammetric survey. In archaeological contexts, the object undergoes an exhaustive examination, entailing the recording of pivotal details such as the catalog number, excavation number, and archaeological nomenclature. This foundational step sets the stage for a comprehensive understanding of the object's historical and archaeological significance. Delving deeper, the material type is scrutinized with a keen eye on reflectivity and reflection patterns, crucial factors guiding the subsequent lighting strategy. For objects with high reflectivity, a judicious choice is made to avoid direct light, showcasing F.O.P.P.A.'s commitment to adapting methodologies based on material characteristics. Opting for the elegance of natural light, the sub-phase proceeds with the measurement of essential object parameters, capturing its essence through a photograph, thoughtfully positioned alongside a meter for scale reference. For structures, the Identification sub-phase takes into account multifaceted considerations, ranging from assessing access hours and public presence to evaluating weather conditions during the survey. In outdoor surveys, a nuanced approach involves adhering to a 15-minute acquisition window to accommodate the dynamic shadow movements induced by the sun—a strategic response to environmental variables. In the controlled setting of a museum, the protocol navigates challenges posed by uncontrollable light entry points and potential disturbances arising from reflective surfaces like glass or mirrors. F.O.P.P.A. embraces a holistic approach, anticipating and addressing diverse challenges across archaeological contexts.

In Database compile those properties: Archaeological Name, Position UTM GPS, Cultural Reference, Absolute Datation

Eidotype

RUNNING EXAMPLE

Eidotype



Eidotype:
Drawn representation of the acquisition maneuvers to be carried out on the specimen taking into account the lighting points, the maneuvering spaces and the type of acquisition that you will choose to adopt, in this case a spiral acquisition.

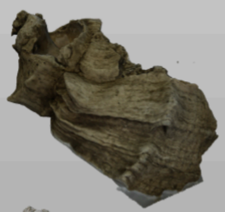
The Eidotype Drawing, is a nuanced blend of logical analysis and graphic representation. When dealing with objects, the process entails a comprehensive examination of the object's general shape, indentations, and potential shadow-prone areas, strategically considering relief generation. This meticulous observation extends to identifying critical components and areas of heightened interest based on the specific research focus. The graphical depiction of the camera's trajectory during the survey becomes a pivotal aspect of the eidotype, offering a strategic overview of the intended acquisition path. For objects, the emphasis lies in capturing their overall morphology, with particular attention to features prone to casting shadows during the relief generation process. Critical and intriguing elements are precisely documented, aligning the eidotype with the objective-driven nature of F.O.P.P.A. Moreover, the eidotype serves as a platform to chart the camera's course during the survey, a visual roadmap that optimizes the survey approach. This graphical representation not only aids in strategizing the acquisition but also initiates the process of determining the most suitable type of acquisition, a decision rooted in the unique characteristics of the surveyed object—be it an archaeological find or a structural element. In essence, the Eidotype Drawing within F.O.P.P.A. transcends a mere illustrative phase; it becomes a strategic precursor, aligning the researcher with the nuances of the object, the environmental conditions, and the intended survey methodology. This integration of logical and visual elements at the outset signifies F.O.P.P.A.'s commitment to a structured, adaptable, and objective-driven approach in photogrammetric practices.

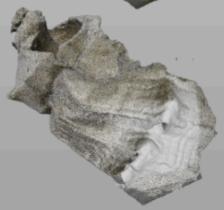
In Database upload media: Picture of Eidotype

Mask

RUNNING EXAMPLE

Mask





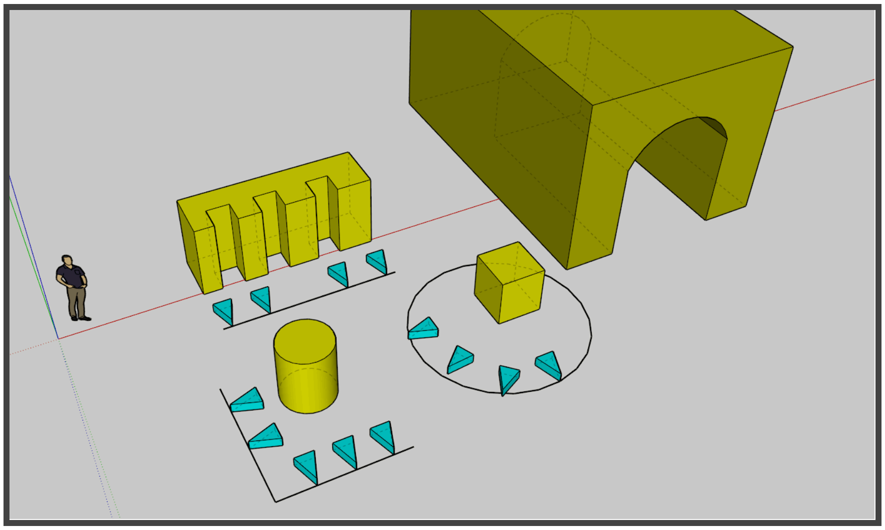
Mask:
Test acquisition called mask, in this case the shell was used as a reference for all the findings acquired:
study of the table background (white, in contrast with the colors of the finds)
insertion of a meter with known dimensions to be inserted in the acquisition

The Mask Relief sub-phase within the F.O.P.P.A. protocol represents a stage characterized by a superficial and swift survey, strategically designed to examine the camera's interaction with light sources. This distinctive facet isn't intended as a comprehensive survey but rather focuses on understanding how the camera responds to different lighting scenarios. In the realm of archaeological finds, it's important to note that generating a mask for each individual artifact isn't obligatory. Instead, emphasis is placed on studying light behavior within the prepared set. This strategic approach optimizes efficiency without compromising precision, aligning with F.O.P.P.A.'s commitment to streamlined yet effective methodologies.
Crucially, this sub-phase involves determining the placement of a solid object with known measurements within the survey context. This object serves a dual purpose: firstly, as a reference point for accurate measurements, and secondly, as a meticulous tool for analyzing potential deformities that might be challenging to discern on the surveyed exhibit. The choice of this safety object is deliberate, favoring monochromaticity, a color absent on the surveyed object, and a linear, well-defined shape, often exemplified by a cube. In scenarios involving the survey of big structures where multiple surveys may be requisite, the Mask Relief sub-phase assumes an additional role. It strives to compile a comprehensive survey of the entire structure, offering a preliminary reference point. This holistic approach facilitates subsequent alignment processes, ensuring the coherence of various point clouds and identifying potential discrepancies or errors. The moment of generation of the machine also represents a fundamental moment of testing of the entire production chain of the 3D model. In fact, for the creation of the Mask a survey is actually carried out, which, although rough and superficial, allows us to go through all the various subsequent phases. The acquisition in this context is free but for reasons of simplicity we suggest a freehand spiral acquisition (where applicable). At this point we proceed with entering the photographs into the Processing software to subject them to Preliminary Processing. We suggest the use of a generic alignment algorithm or setting, a similar approach to adopt for densification and mesh creation. At this point you export the 3D model without any modeling process and that will be the Reference Mask.

In Database upload media: upload the 3d object of the mask, in the case of small findings just one mask for study of the acquisition set.

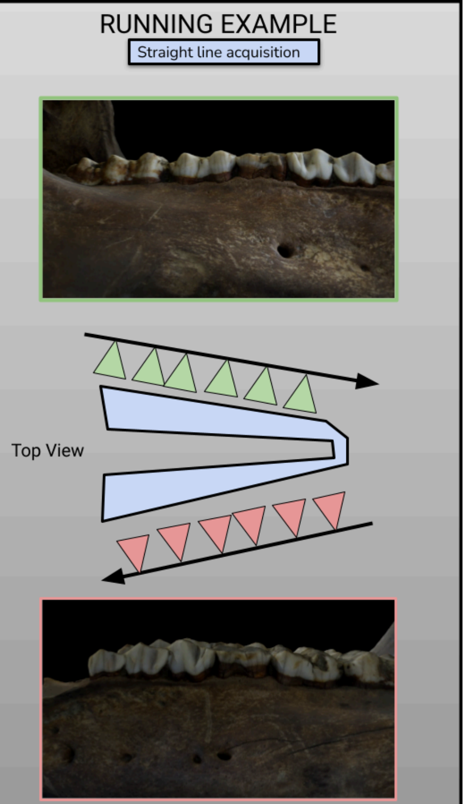
Survey

The Survey sub-phase within the F.O.P.P.A. protocol represents the moment of conducting the actual survey. The number of photographs taken is contingent on the availability of software and hardware. For artifacts, light placement is determined by considerations from preceding sub-phases. Throughout the survey, it is crucial to avoid any direct light points in the camera's view, preventing potential color information "burnout." Notably, any adjustments to the gamma curve during the survey alter the colorimetric relationships between photographs, hindering the identification of color points across images. Photo superimposition is fundamental, irrespective of the chosen acquisition technique, with a recommended alignment value falling between 65% and 80%. Essential precautions include refraining from using zoom during acquisition and avoiding any alterations to the lens distortion ratio. In the realm of artifact acquisition, the necessity of constructing a set isn't universal. If the survey environment boasts an array of color points not aligned with the artifact, they can enhance relief accuracy. Consequently, set construction should be avoided where unnecessary, preventing concentration of information solely on the specimen and risking oversight of potential deformities. Selection of the acquisition technique adheres to the "rule of opposites." This entails identifying the primary, essential shape of the object (cube, parallelepiped, sphere, cylinder, plane, etc.) and employing an acquisition method that counters it. This strategy mitigates the risk of hindering the machine's ability to recognize color point distribution transformations across photographs, ensuring a more accurate representation of the object's relief.



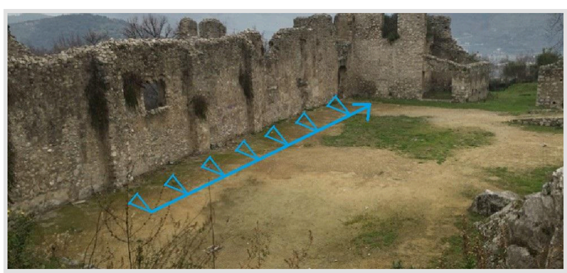
Schematic representation of the "Rule of Opposites"

Straight line acquisition



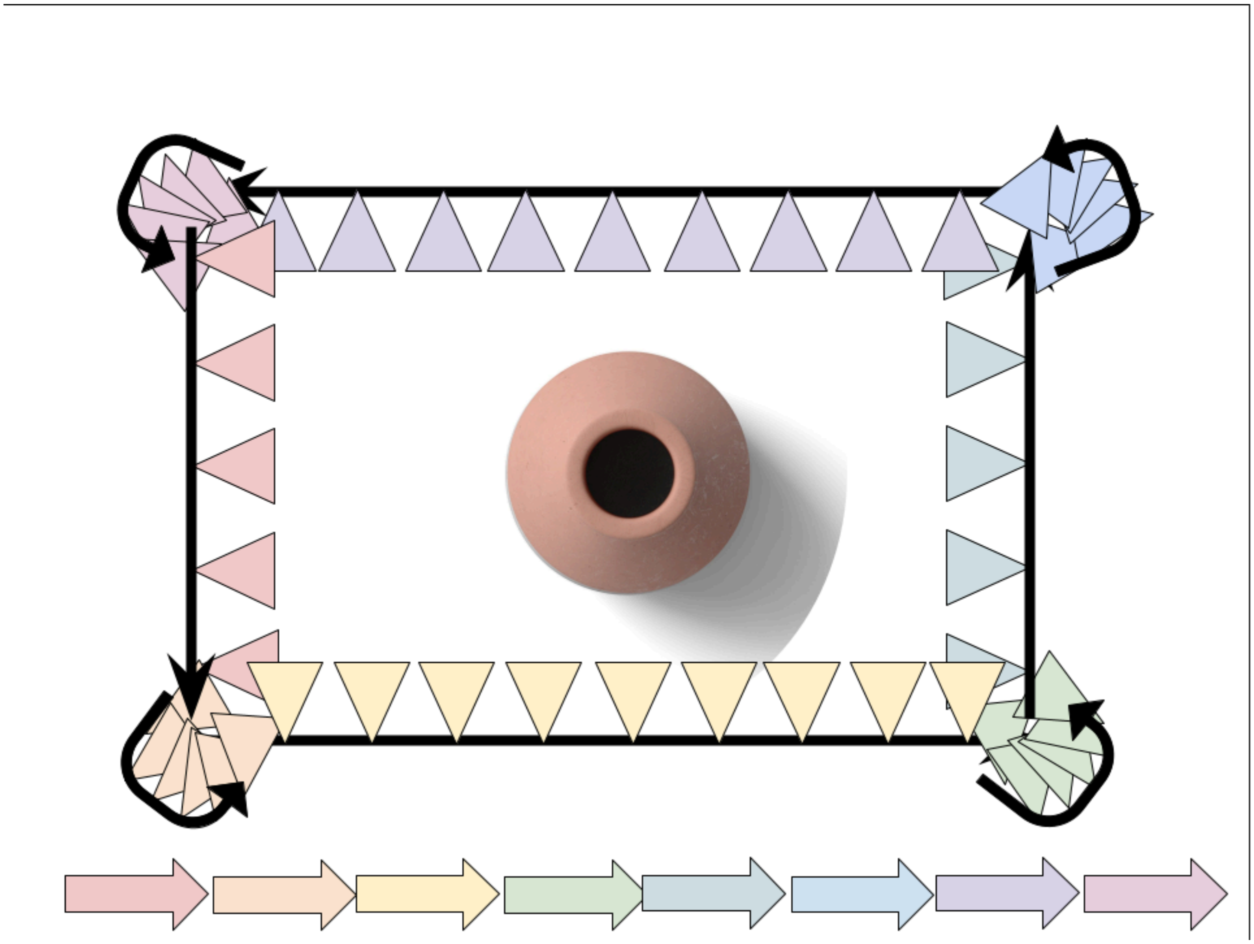
Straight Line Acquisition:
Example of linear acquisition applied to the find under examination, in this case applied to the lateral walls of the find

The Straight Line Acquisition method in the F.O.P.P.A. protocol is tailored for large buildings and facades, offering an efficient approach to surveying expansive structures. This technique involves moving the camera along a straight line parallel to the structure while turning it perpendicular to the object of the survey. As the camera progresses, the focal point of the relief moves in parallel along the structure, maintaining an orientation perpendicular to the object being surveyed. This approach is particularly suitable for capturing extensive architectural elements and intricate details present in large structures. The methodology ensures comprehensive coverage of the structure, minimizing the risk of overlooking critical aspects during the survey. An essential criterion for the success of Straight Line Acquisition is maintaining a significant overlap between successive photographs, set at no less than 70%. This substantial overlap is crucial for subsequent data processing and analysis, facilitating accurate stitching and alignment of images to reconstruct a cohesive and detailed 3D model of the structure.

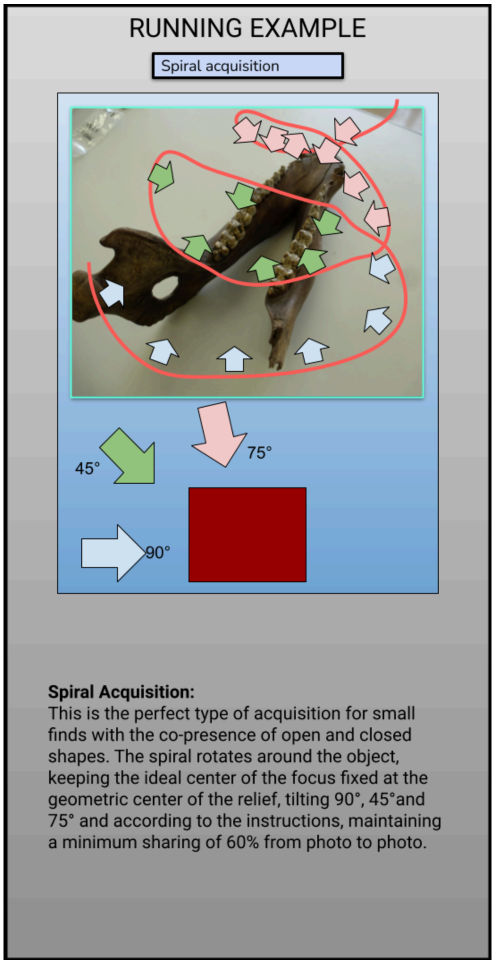


In Database compile those properties: Type of Acquisition

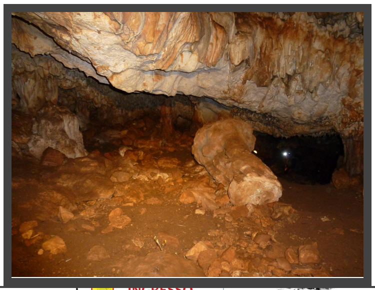
In the context of linear acquisition applied to objects, it must be taken into account that the moment of transition along the corners that connect the various segments of the acquisition strip require a number of photos equal to or greater than that of the linear stripes proper and that define the segments of the ideal square surrounding the find. This overabundance is required in part because the relief, relying more on color points and less on perspective projections.

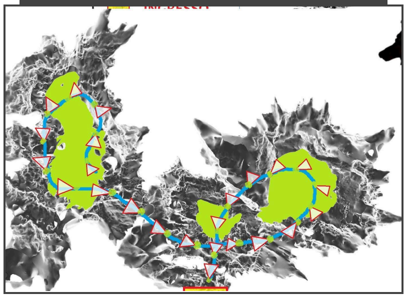


Spiral acquisition



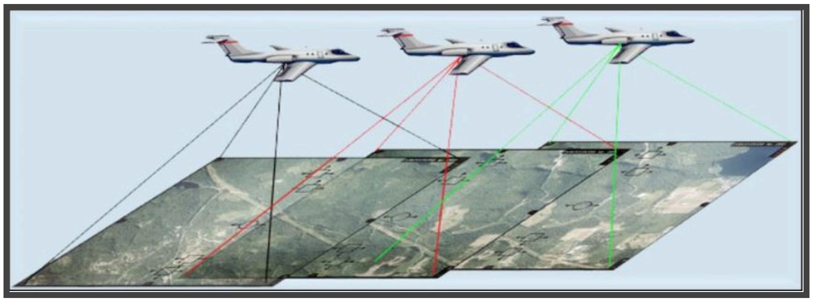
The Spiral Acquisition method within the F.O.P.P.A. protocol offers a systematic approach to capturing detailed information about individual finds and artifacts. This technique involves defining an ideal focal point at the center of the object under survey, directing the camera towards this center, and executing a comprehensive acquisition of photographs. The process unfolds initially with perpendicular shots, followed by an inclination of 45°, and finally at an inclination of 75°. This approach is particularly effective for documenting single finds and artifacts, ensuring that each aspect is thoroughly captured from different perspectives. The spiral acquisition technique becomes invaluable when intricate details or features of an object need to be emphasized. Beyond its application for individual finds, the spiral acquisition method can also be adapted for data collection in cavities or structures resembling corridors. In such instances, the focal length is strategically positioned at the bottom, and the base of the spiral is aligned perpendicular to the ground.





In Database compile those properties: Type of Acquisition

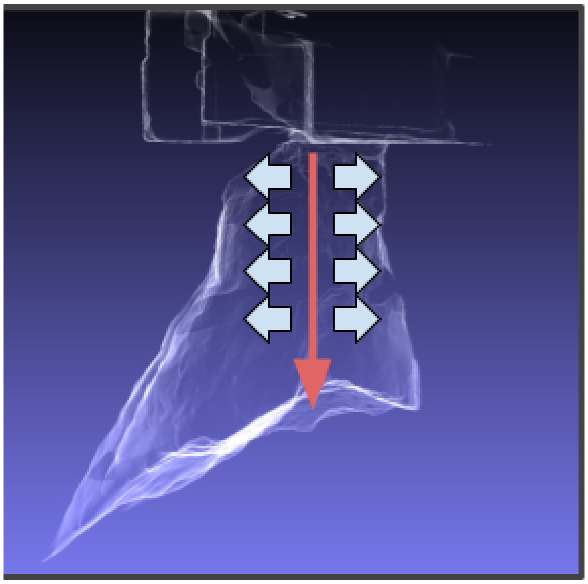
Aerial acquisition

The Aerial Acquisition technique, a specialized variant of straight-line acquisition, involves moving parallel to the ground while tilting the camera downward. This approach, aptly named for its suitability with drones in territorial surveys, also proves exceptionally effective for documenting archaeological trenches. This method provides a unique perspective by capturing images from an elevated position, offering a comprehensive overview of the surveyed area. Particularly advantageous when used with a drone, the aerial acquisition technique facilitates efficient and detailed data collection for extensive territories. The drone's mobility allows for flexible maneuvering, ensuring optimal coverage of the archaeological site. In the context of archaeological trenches, the aerial acquisition technique excels in providing a bird's-eye view of the excavation area. This vantage point offers valuable insights into the spatial layout, features, and relationships within the trenches.
 
In Database compile those properties: Type of Acquisition

Free vertical or with a drone or with pole

Vertical capture, a nuanced application of straight-line acquisition, is specifically designed for elevations that extend significantly in height. In instances where the surveyed structure reaches considerable heights, the use of a drone becomes highly recommended for optimal results. This technique provides a systematic approach to capturing images along a vertical plane, ensuring a comprehensive documentation of tall structures.

Drones, equipped with cameras or even mobile phones, serve as ideal tools for executing vertical acquisitions, offering unparalleled flexibility and efficiency. The choice of the instrument largely depends on the specific requirements of the survey. The F.O.P.P.A. protocol recognizes the importance of accommodating diverse scenarios, allowing practitioners to employ different devices based on the nature of the archaeological site and the objectives of the survey. This category also includes vertically developed surveys obtained by raising the camera using a rod or pole
Vertical acquisition is particularly advantageous when applied to structures with towering heights, such as architectural elements, monuments, or stratigraphic sections.

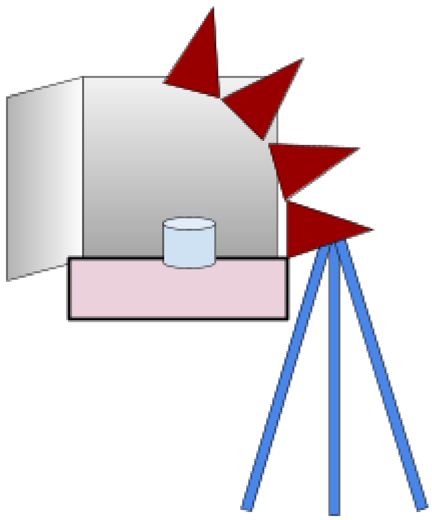


In Database compile those properties: Type of Acquisition

Spiral with rotating base

The survey on a rotating base is actually a direct derivation of the spiral survey where it is not the camera that moves around the object but the movement is simulated by rotating the object on a rotating base. In order for this "deception" of the elaboration of the structure from the movement to work, it is necessary that the set is completely neutral and dominated by a single color that is absent from the object under examination. Furthermore, it is necessary to create a background with a set that contains the reflection and reflectance phenomena of the detected object. To do this, for example, with objects with high reflectivity you will have to use a background made of an extremely absorbent material such as velvet.

The stripes, i.e. the position of the camera in reference to the object, must be four in number with an inclination of 0 degrees (perpendicular to the object), 15 degrees, 45 degrees and 75 degrees. The rotating base, between each photograph, must rotate from a minimum of 5 degrees up to a maximum of 15 degrees, thus also determining the number of photographs. Given the breadth of the topic, fixed-base acquisition is partially covered and explored in depth in appendix #2 of the Foppa manual.

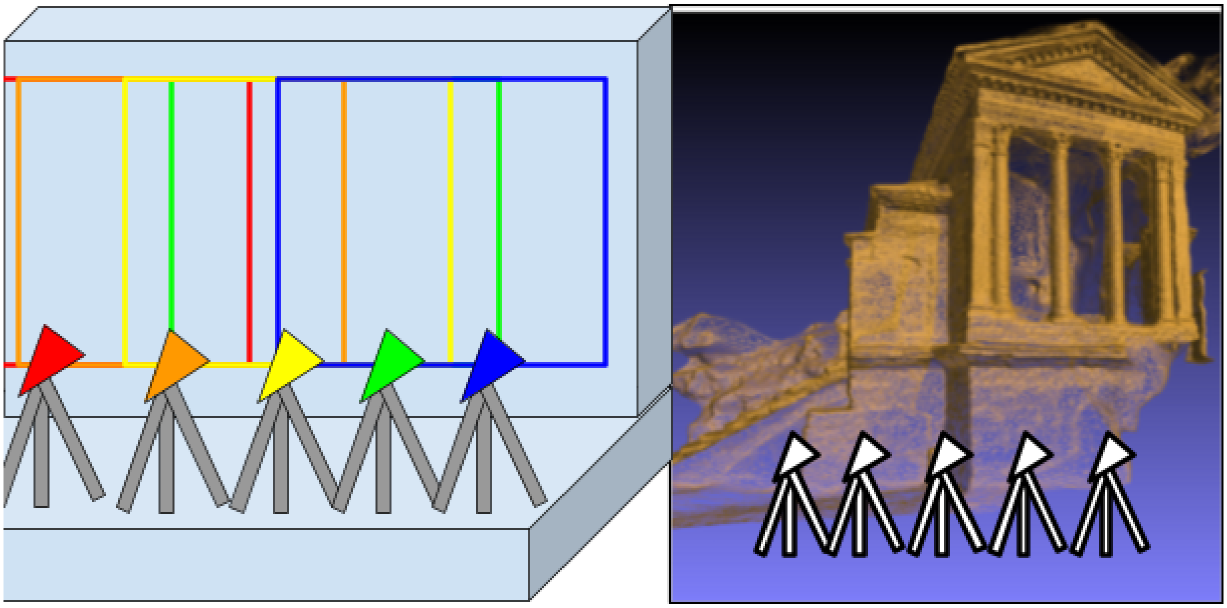


In Database compile those properties: Type of Acquisition

Easel linear

Acquisition with a camera fixed on a tripod is an alternative freehand linear acquisition where the camera is secured to a tripod and the lateral movements necessary to carry out the acquisitions are carried out according to a pre-established order, punctual and at equidistant points, placing the camera perpendicular to the the object of the survey

This type of acquisition is useful in wall analysis contexts where the extruded or protruding elements are extremely small but colorimetrically extremely recognizable (a frescoed wall with cracks for example) and is to be avoided where there is little variation in the color range.
Each photo must share at least 70% of the elements with the photo that precedes it, therefore requiring numerous acquisitions.



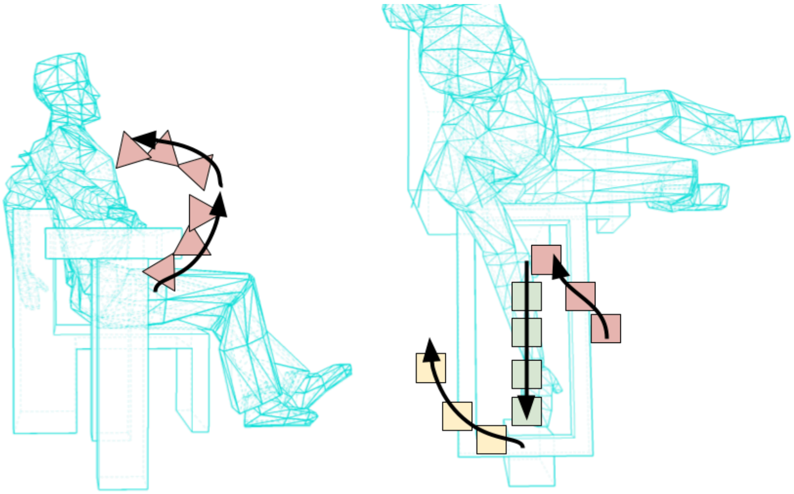
In Database compile those properties: Type of Acquisition

Structured mixed

In specific contexts it may be necessary not to exclusively use a linear, spiral or vertical relief but a combination of them. The structured survey, unlike the unstructured one, is however made up of survey parts which respect the reference parameters of the various modules that compose it. This means that a combination of spiral survey, linear survey and then spiral survey still ensures that the characteristic parameters of the various types of acquisition are respected.

When possible it is better to avoid mixed acquisitions structured especially for large structures and prefer differentiated acquisitions for the various areas of the structure, on the contrary this type of acquisition can be functional for extremely irregular small objects.

This type of acquisition is also extremely efficient for the acquisition of body parts. In this regard, a specific in-depth analysis is indicated for the survey of parts of the human body indicated in Appendix #1 of the FOPPA Manual.



In Database compile those properties: Type of Acquisition

Mixed unstructured

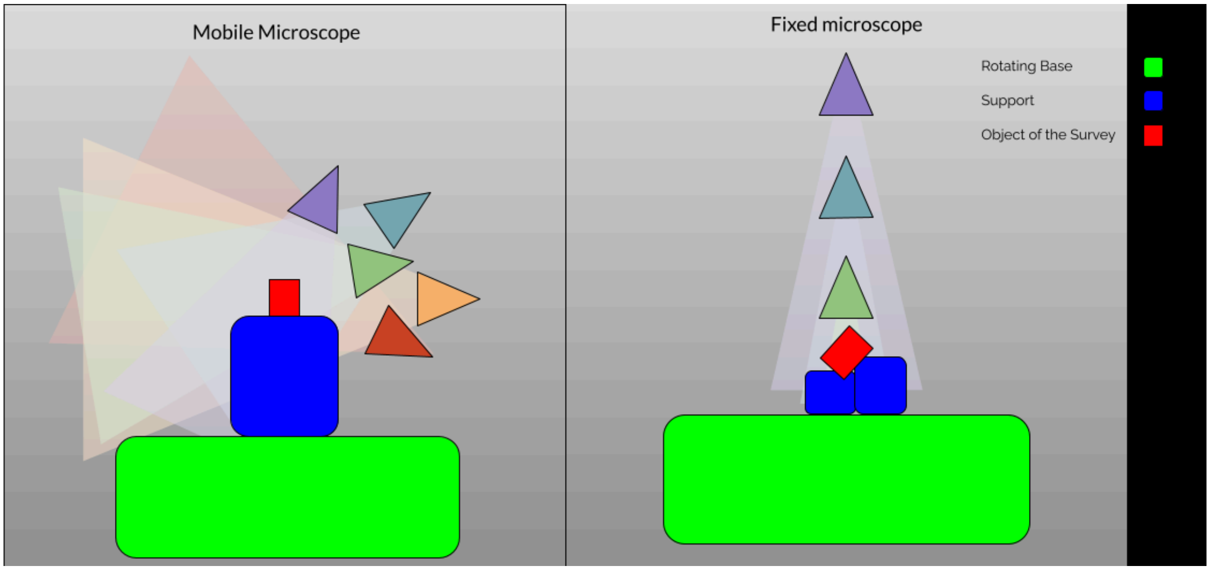
An unstructured mixed acquisition indicates an acquisition whose composition cannot be traced back to the acquisition modules described up to this point. This type of acquisition may be necessary for the acquisition of walkable environments or landscape elements characterized by extreme irregularity. This type of acquisition should always be avoided where possible and the Mixed Structured one should always be preferred, looking at where the modules can be applied along the acquisition path. Where this is not possible, as in the case of extracting frames from a video not originally designed for acquisition, it is necessary to focus attention on sharing the color points between the various photographs (not being able to rely on the information coming from stereoscopy as it is not particularly characterized by a structured acquisition module).
For this reason the photos must share no less than 80% of the representation between the various photographs, without changing the vertical/horizontal orientation of the camera. This system can be used for the acquisition of deep tunnels characterized by narrow spaces where it is not possible to move with structured acquisitions.

In these cases it is necessary to provide a large amount of information on the color points, which is why it is necessary to take a large number of photographs, many more than would be necessary to take in other contexts.

In Database compile those properties: Type of Acquisition

Rotating base microscope

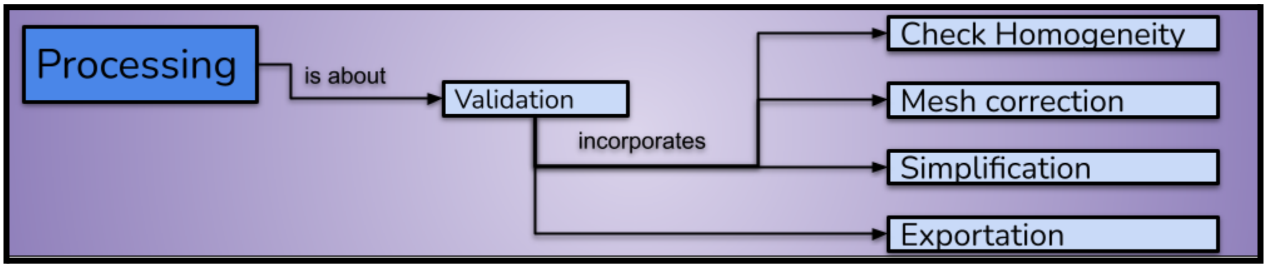
Rotating base acquisitions carried out with the aid of a microscope as a camera are specific for the acquisition of small or tiny objects (i.e. within a diameter range of less than 2cm) and require a series of specific measures aimed at overcoming a series of obstacles such as the presence of an orthographic lens (a characteristic of microscopes) or the management of acquisition points marked with greater rigor. Due to the vastness of the topic, the entire Appendix #2 of the FOPPA Protocol is dedicated to the acquisition of tiny objects.



In Database compile those properties: Type of Acquisition

Processing

The Processing phase of the F.O.P.P.A. protocol marks a stage where the raw data, comprising a multitude of photographs obtained during the survey, undergoes intricate digital transformations to manifest as a coherent and accurate 3D model. This phase, crucial for the fidelity of the final output, unfolds in a series of meticulously orchestrated steps. The initial step involves loading the acquired photographs into photogrammetry software. This software serves as the digital darkroom where the magic of reconstructing a three-dimensional representation from two-dimensional images takes place. The software meticulously analyzes each photograph, identifying common color points across the images to establish correspondences. This process, known as matching, essentially stitches together the photographic puzzle, creating a unified point cloud that encapsulates the spatial relationships between the photographed elements. Following matching, the point cloud undergoes a densification process. This step enhances the resolution and precision of the point cloud, filling in gaps and refining details. Densification is crucial for ensuring the accuracy of the final 3D model by capturing intricate features and nuances present in the original object or structure. The next milestone in the Processing phase is the reconstruction of the 3D model mesh. This involves connecting the dots, quite literally, as the points in the densified cloud are linked to create a mesh that accurately represents the geometry of the surveyed subject. The mesh, resembling a digital skeleton, serves as the framework for the final 3D model. Simultaneously, textures are assigned to the mesh, bringing the model to life with realistic surface details. Texturing involves mapping the colors and patterns from the original photographs onto the corresponding areas of the 3D model, creating a visually authentic representation. It's noteworthy that the choice of algorithms for matching, densification, and mesh creation is a critical consideration, and it varies among photogrammetry software. Different software packages employ distinct methodologies, and practitioners must align these choices with the specific type of acquisition conducted during the survey. Ensuring consistency between the selected algorithms and the acquisition type is paramount for maintaining the integrity and accuracy of the reconstructed 3D model.



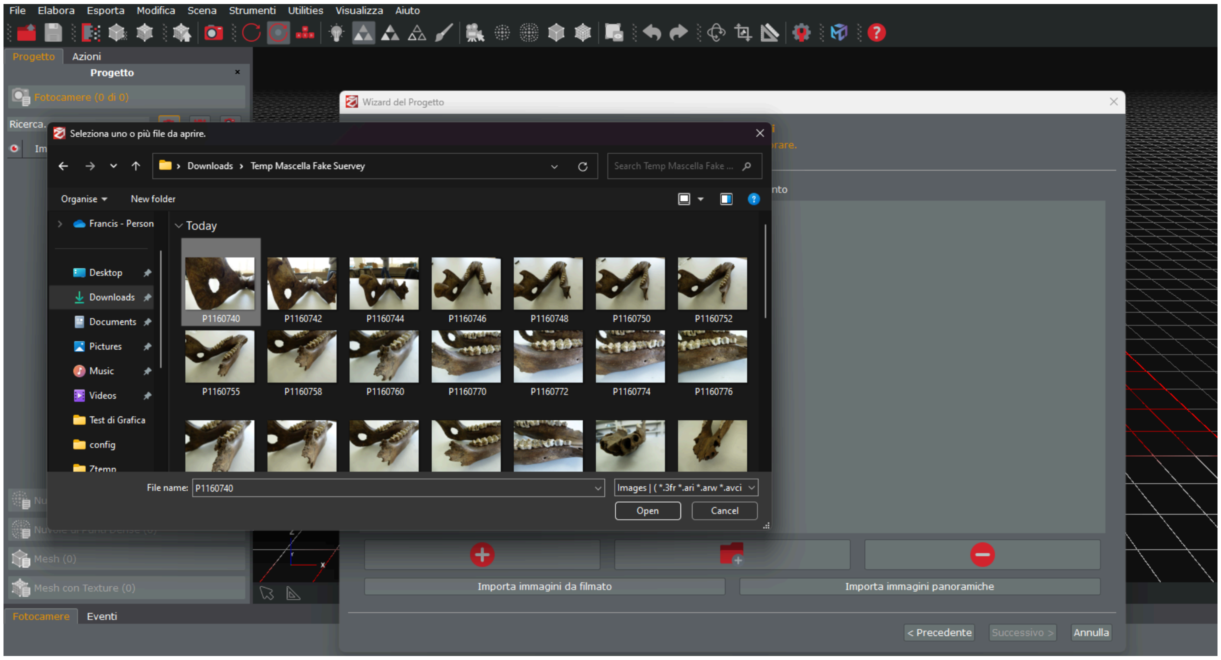
Processing example

Each processing software is different but each has common mandatory phases. Here we report as an example an elaboration of the finding used as a Running Example through the Zephyr software.

Regardless of the software used, each processing consists of the following phases:

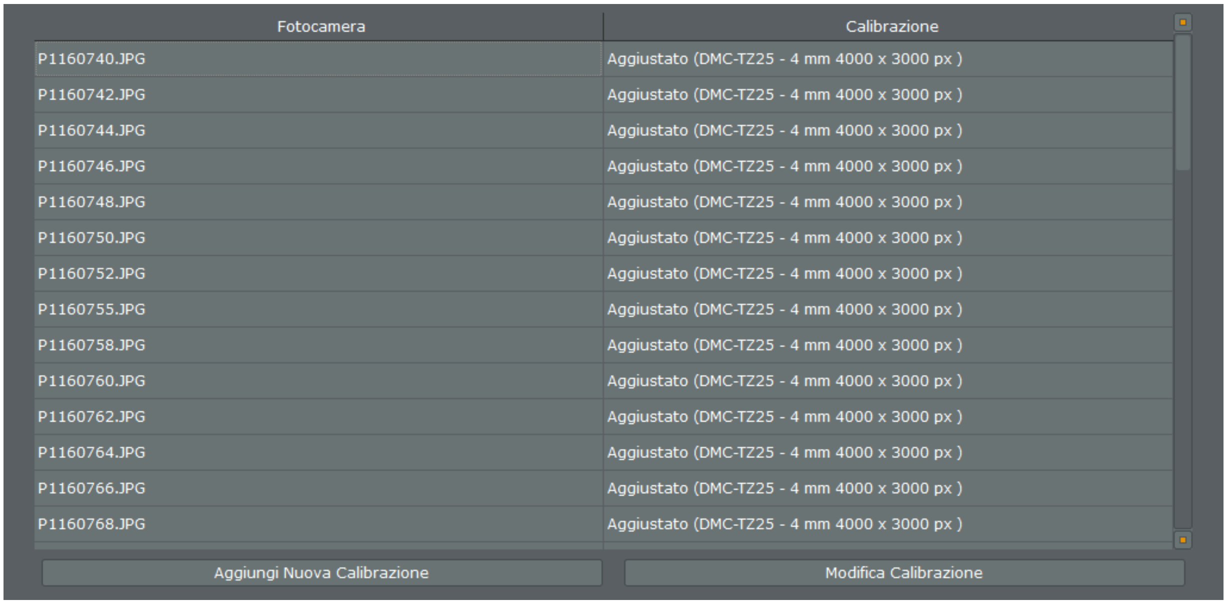
1. Insertion of the collected photographs,
2. calibration and correction of lens distortion starting from the library of lenses present in the software,
3. matching of the points shared between the various photos,
4. densification of the point cloud,
5. creation of the surface known as mesh,
6. addition of textures taken from the photograph.

1



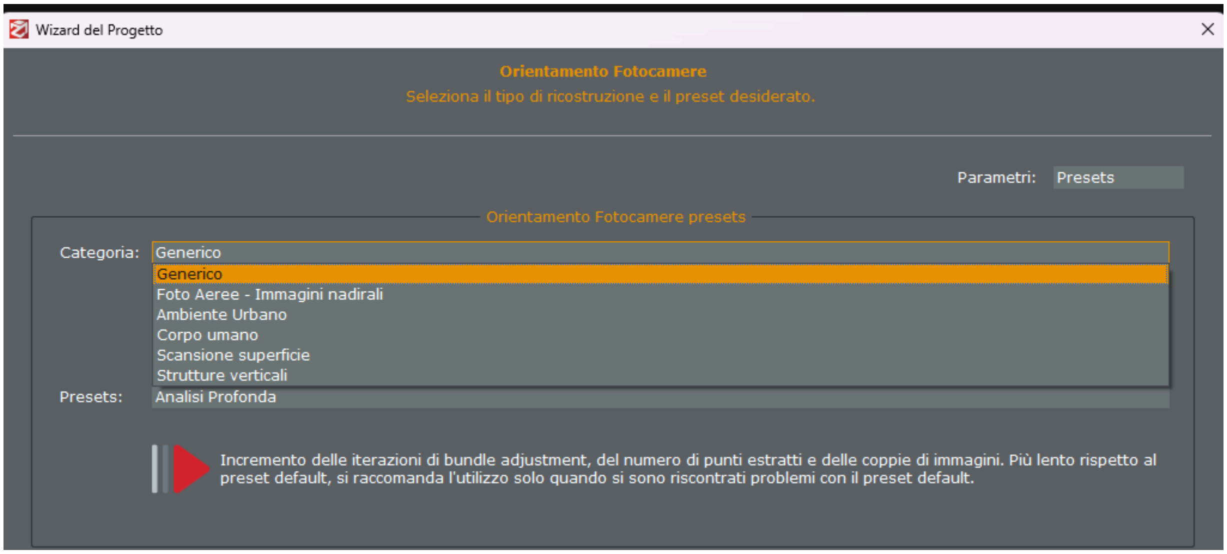
Insertion of collected photos

2



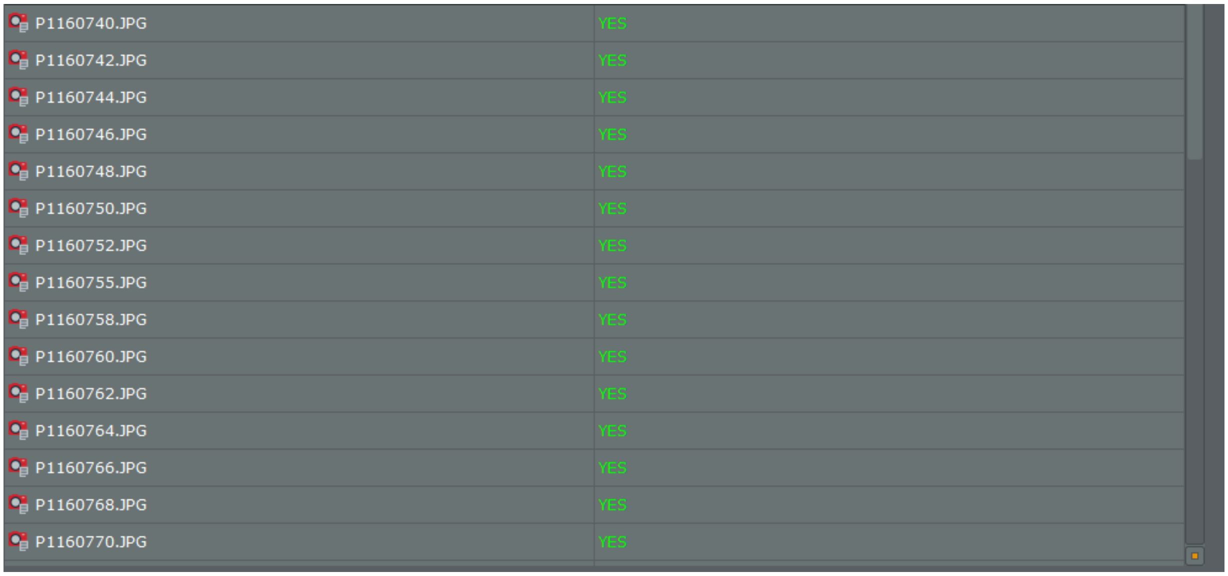
Each software corrects the deformation error intrinsic to each photographic lens and can do this thanks to a library it has inside which allows it to recognize the lens body.

3



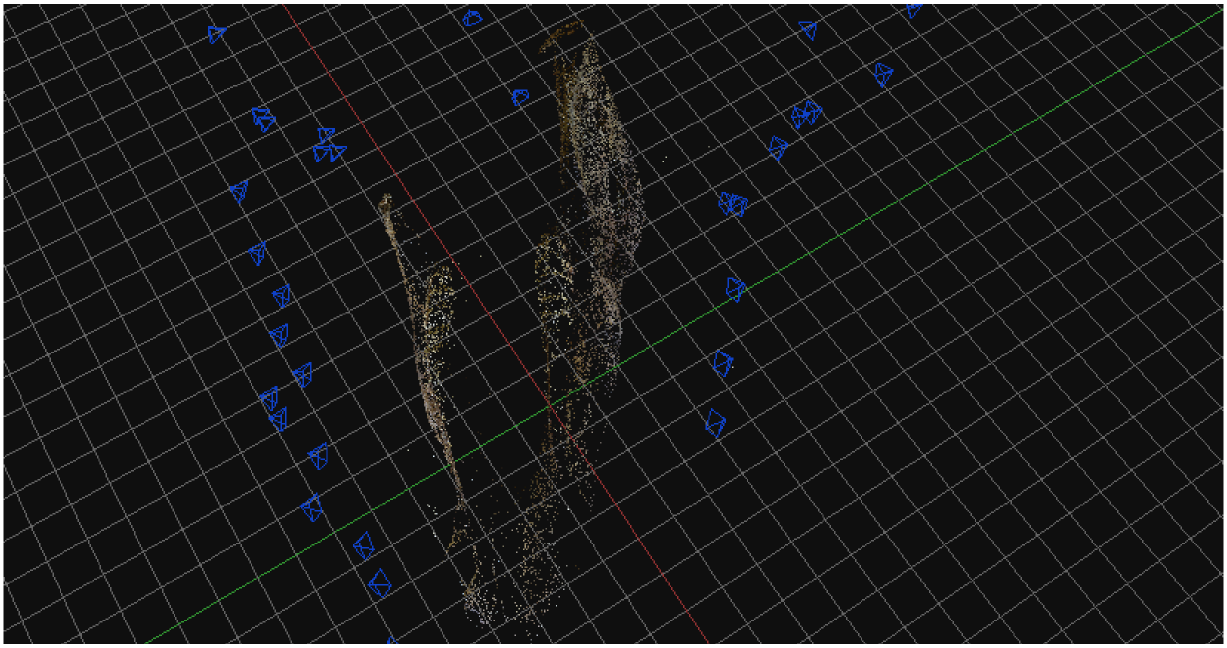
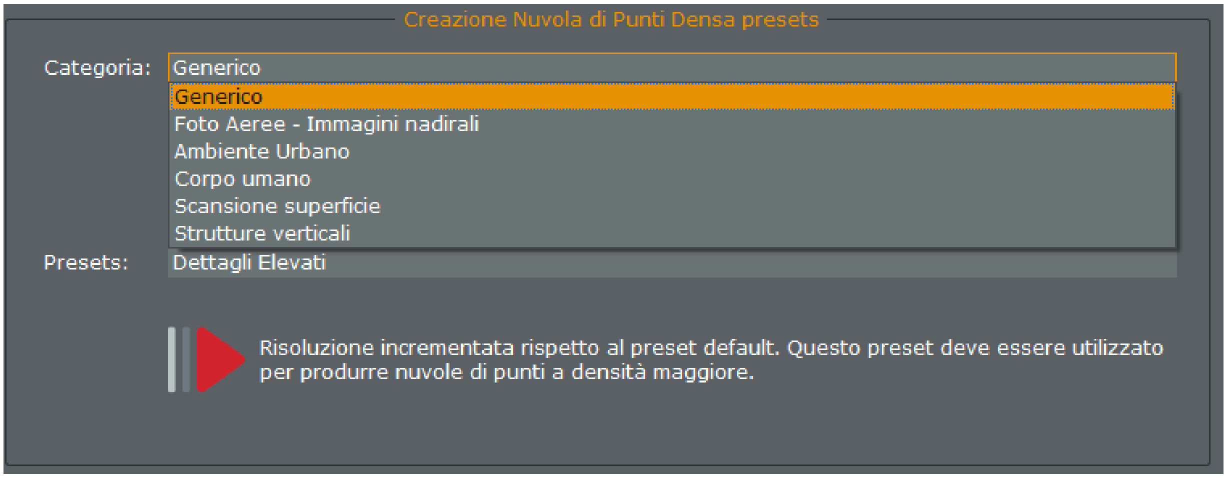
Some software indicates exactly which algorithms will be used for matching, others instead give a preset that refers to the nature of the object. For this reason it is necessary to keep in mind what type of acquisition took place and maintain a consistent approach also in choosing the matching method. For example, the spiral survey can be considered the "Standard" acquisition system and is consistent with the "Generic" zephyr preset. To orient yourself in the choice, go to the chapter on acquisition methods and choose based on the type of object in which this technique is most used, placing the emphasis for the final choice not so much on the real nature of the object but on the acquisition method used.

3.1



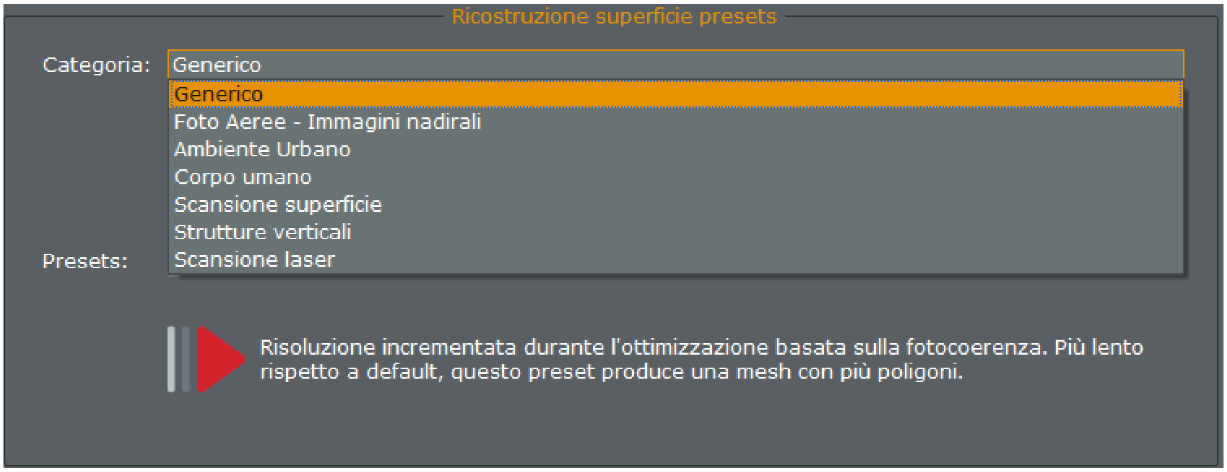
P1160740.JPG	YES
P1160742.JPG	YES
P1160744.JPG	YES
P1160746.JPG	YES
P1160748.JPG	YES
P1160750.JPG	YES
P1160752.JPG	YES
P1160755.JPG	YES
P1160758.JPG	YES
P1160760.JPG	YES
P1160762.JPG	YES
P1160764.JPG	YES
P1160766.JPG	YES
P1160768.JPG	YES
P1160770.JPG	YES

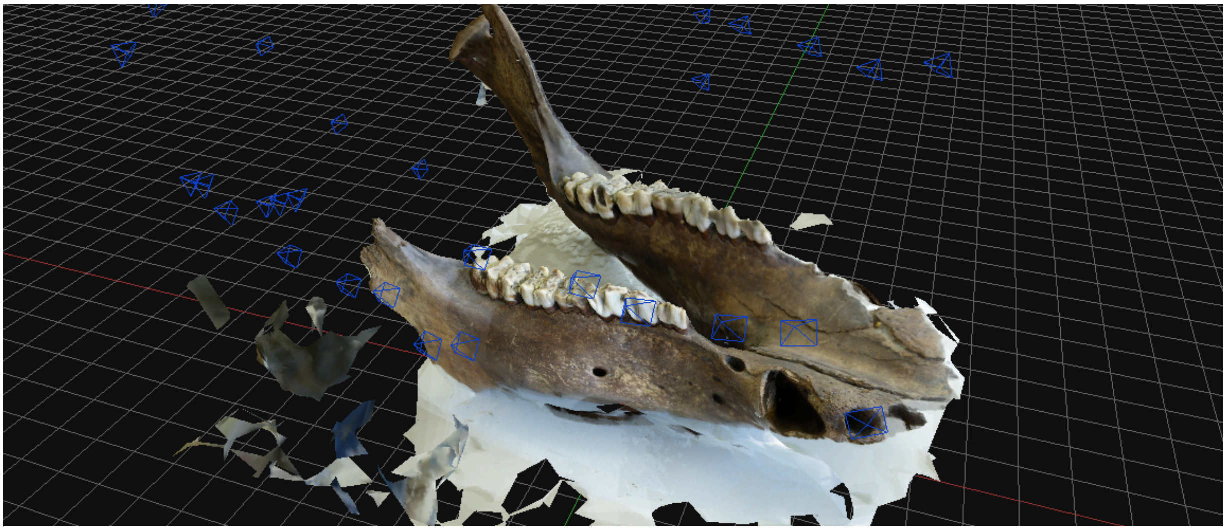
Whenever possible, we must prefer software that is based on local processing and therefore does not send the photographs to a third-party server and then return the finished model. In the context of local processing software, a list of photographs is provided in which the color points in common between the various photographs have been successfully recognized: if the number of photographs recognized is less than 80% overall, the survey must be repeated.

3.2

The result is the first point cloud. This three-dimensional object is the true survey and it must be kept in mind that all subsequent operations are in fact a necessary falsification of the process.
4

For densification, as well as in the subsequent creation of the surface, where the software, as in the case of Zephyr 3D, gives the possibility to choose the reconstruction method, one must try where possible to respect, as in the case of matching, the method of acquisition of the photographs . Densification, in the context of point cloud processing, refers to the

augmentation of data points to achieve a higher density in a given area. It involves interpolating or adding points between existing ones, enhancing the level of detail and precision in the point cloud representation. Densification is a crucial step in refining the accuracy and completeness of a 3D model generated from the point cloud. This process aids in capturing intricate details, irregular surfaces, and subtle features by ensuring that the spatial distribution of points accurately reflects the geometry of the surveyed object or environment.

5

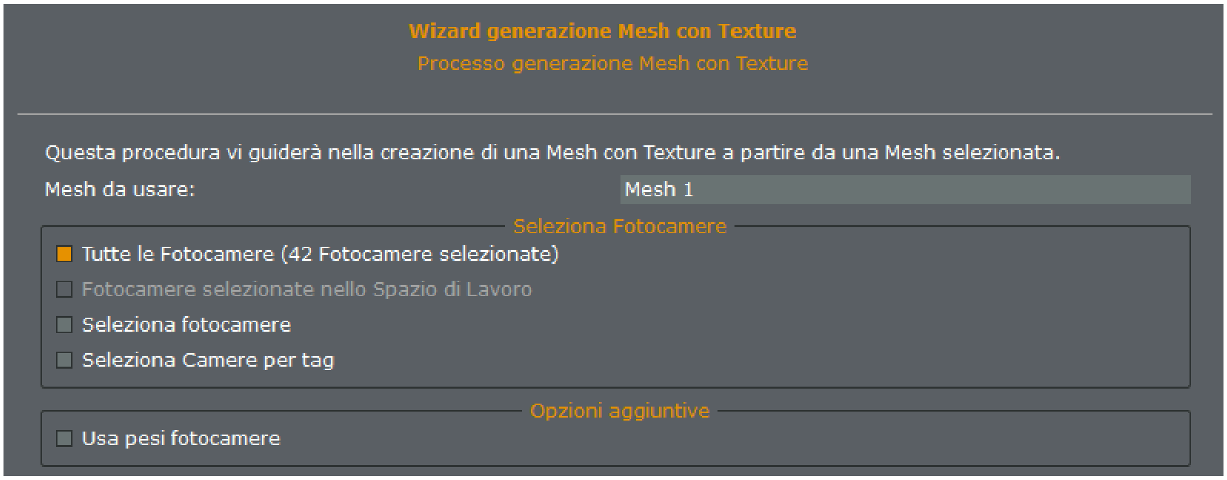




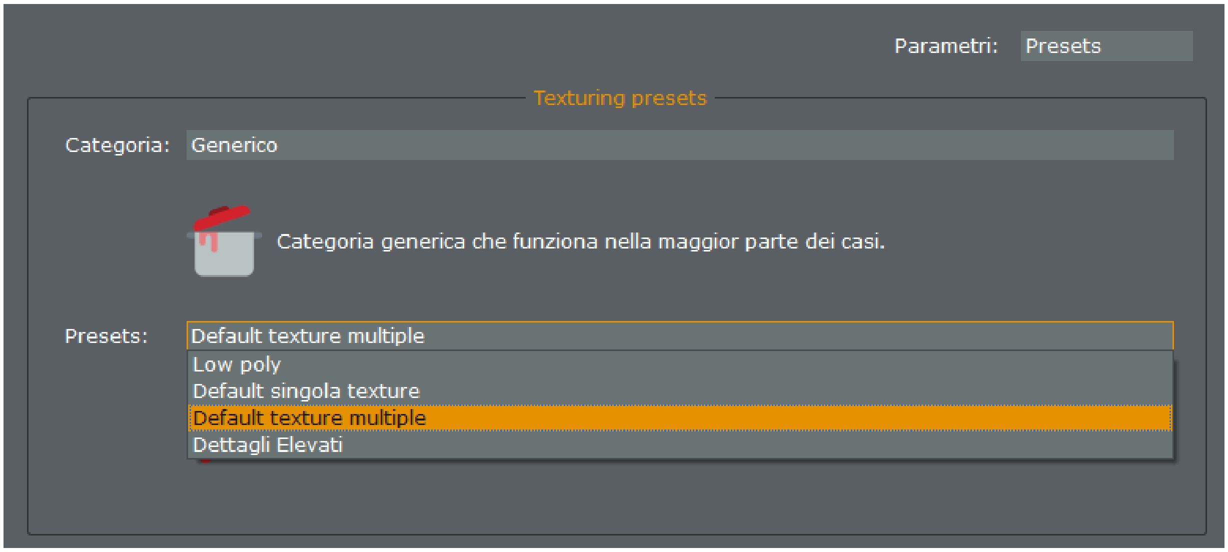
The creation of the mesh, or surface, in point cloud processing involves converting the dense cloud of data points into a continuous, three-dimensional representation of the surveyed object or environment. This process is critical for generating a tangible and visually realistic

3D model. Initially, algorithms analyze the relationships between adjacent points, forming triangles that collectively compose the mesh surface. The density and arrangement of these triangles define the model's resolution and detail. To enhance the accuracy of the representation, software algorithms often incorporate various interpolation and smoothing techniques. The resultant mesh effectively encapsulates the geometry and topology of the original subject, allowing for further analysis, visualization, and interaction.

6



Texture creation can be single or multiple. In this case it is good to keep in mind what the purposes of the survey are: for example, if the survey is aimed at a detailed analysis of a find or a large structure it is best to opt for multiple textures, but if there are problems inherent to the storage of the 3D model or database does not support multiple texture files, it is best to proceed with a single texture. Within a project it is good to choose an approach and keep it for all the objects of the survey in order to establish internal coherence within the project.



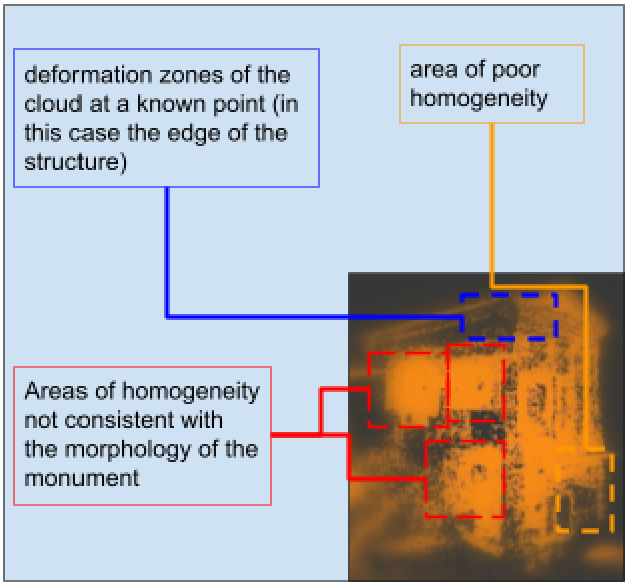
In Database compile those properties: Processing Software & Motivation Choice

Validation

The Validation Sub-Phase within the F.O.P.P.A. protocol is a critical stage ensuring the accuracy and reliability of the 3D model. In this step, the fidelity of measurements is rigorously verified through meticulous controls, aimed at confirming the precision of the acquired data. A fundamental aspect of validation involves comparing the surveyed object with a reference object of known measurements, strategically placed during the acquisition. This reference object serves as a benchmark, allowing for the verification of any deformations or inaccuracies in the shape of the surveyed object. By ensuring a seamless alignment between the reference object and the acquired model, the protocol establishes the correctness of the measurements. Furthermore, validation extends to the scrutiny of the relationship between texture and mesh. This process involves assessing the coherence and alignment of the surface textures mapped onto the 3D model. Any discrepancies or distortions between the texture and mesh are carefully examined and corrected to guarantee a seamless integration of visual details. Another facet of the validation process is the examination of the laboratory set, emphasizing the importance of avoiding black sets whenever possible. The accurate detection of the laboratory set is crucial for the overall quality of the 3D model. By verifying the correct recognition of the set, the protocol ensures that potential errors or artifacts introduced during the acquisition or processing phases are promptly identified and addressed.
In Database compile those properties: Validation Information. In the property description, enter the following information. 1)“n photos 2)algoritm/form used for match 3)algoritm/form used for densification 4) alg/for mesh reconstruction 5)texturing type

Check Homogeneity

The Check Homogeneity Sub-Phase in the F.O.P.P.A. protocol serves as a critical assessment to ensure the overall consistency and integrity of the point cloud generated during the processing phase. This meticulous inspection is pivotal in identifying and rectifying any potential deformations or anomalies that might have been introduced in the course of data processing. Homogeneity control involves scrutinizing the point cloud for any irregularities or areas of high densification that deviate from the expected morphological structure of the surveyed object. Anomalies such as arbitrary densifications that do not align with the object's surface or linear densifications that follow the progression of the cameras during acquisition raise red flags. These irregularities could indicate deformations or falsifications introduced during the processing phase and must be thoroughly documented or rectified to maintain the accuracy and authenticity of the 3D model. The objective is to ensure that the point cloud maintains a coherent and uniform structure, aligning with the true morphology of the surveyed object. Inconsistencies in densification patterns are carefully examined, and any deviations from the expected surface structure are investigated. This vigilant scrutiny helps in identifying potential artifacts, distortions, or errors that might compromise the fidelity of the 3D model.



In Database compile those properties: Validation Information. In the property description, enter the following information: 6) homogeneity check

Mesh correction

The Mesh Correction Sub-Phase in the F.O.P.P.A. protocol marks a crucial step in ensuring the integrity and accuracy of the 3D model's surface representation. This meticulous process involves a comprehensive examination of the mesh, verifying its alignment with the true surface morphology of the surveyed object. Verification of the mesh correctness is a multifaceted process, beginning with a reference object of known measurements placed alongside the survey object during acquisition. This ensures that the surface structure of the mesh aligns accurately with the actual geometry of the object. Simultaneously, a careful analysis is conducted to identify areas where the texture and mesh do not coincide. One of the primary challenges addressed during this sub-phase is the distinction between shadows caused by the absence of data, such as areas where the camera was unable to frame, and shadows generated by processing errors. This discrimination is crucial to maintaining the fidelity of the 3D model and ensuring that anomalies or inaccuracies are promptly identified. Additionally, the sub-phase includes a meticulous examination of the average trend of the mesh detail, assessing its consistency with the densification processes applied during the earlier stages. Any deviations from the expected trend are thoroughly investigated, and if necessary, corrective measures are taken. It is imperative to note that any manipulation of the mesh during the survey for cultural heritage is strictly prohibited. Therefore, the definition of every aspect related to the mesh, including corrections and recalculations, must be established during the processing phase. This stringent approach is crucial to preserving the authenticity and scientific rigor of the 3D model, making the Mesh Control Sub-Phase an indispensable component of the F.O.P.P.A. protocol for reliable archaeological documentation.
In Database compile those properties: Validation Information: to the extent that changes have been made to the mesh it is necessary to enter the details of the changes in the notes of this property

Simplification

The Simplification Sub-Phase in the F.O.P.P.A. protocol plays a pivotal role in ensuring the optimal handling and storage of 3D models. This process becomes relevant when the produced 3D model exhibits excessive complexity, making it cumbersome for database integration or virtual museum display. Ideally, it is recommended to plan the survey with the final destination in mind, considering potential simplifications during the processing phase itself. The goal is to maintain the model's fidelity while ensuring it aligns with the specific requirements of the intended platform. However, if the complexity exceeds the manageable threshold, simplification becomes a necessary step. It's crucial to emphasize that this simplification should occur exclusively during the processing phase and never in the modeling phase. The 3D model should not be exported and then simplified, as this risks arbitrary and non-linear damage to the mesh. The simplification process involves employing decimation processes within the processing software. This strategic approach ensures that the densified point cloud is streamlined without compromising the essential points derived from the matching process. By conducting simplification at this stage, the integrity of the model is preserved, and potential issues related to mesh damage or non-linearity are mitigated. It is worth noting that any form of manipulation or simplification should be executed with precision and consideration for the ultimate purpose of the 3D model. This adherence to a principled workflow is essential for maintaining the scientific authenticity and accuracy of archaeological documentation within the F.O.P.P.A. protocol.
In Database compile those properties: Validation Information: to the extent that simplification operations have been carried out on the cloud, it is necessary to indicate to what degree of simplification and using which simplification algorithm

Exportation

The Exportation Sub-Phase serves as the concluding step within the Processing Phase of the F.O.P.P.A. protocol, marking the point where the 3D model transitions out of the processing software. This crucial stage determines the format in which the model will be conveyed, necessitating careful consideration to maintain consistency and integrity.

Choosing the appropriate format during exportation is a critical decision, as it is generally preferred to retain the initial format throughout subsequent modeling phases. However, practical constraints or specific requirements of the processing software may, at times, necessitate a format change.

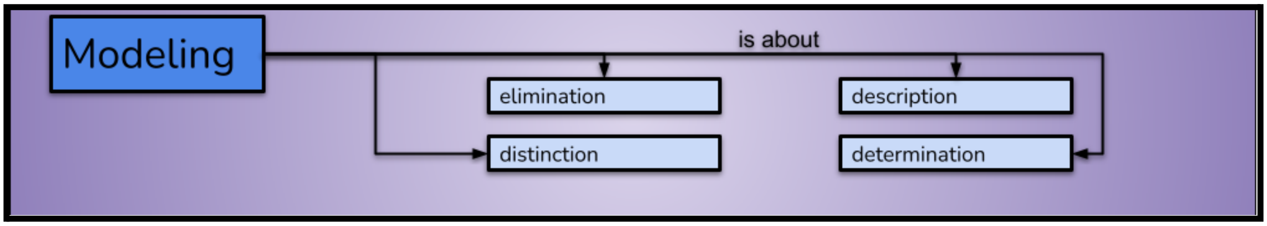
The selection of the export format is influenced by several factors, including the intended use of the 3D model, compatibility with downstream modeling software, and adherence to established standards. The F.O.P.P.A. protocol emphasizes the importance of strategic decision-making during this sub-phase to ensure seamless continuity in the modeling workflow.

While the ideal scenario involves maintaining the format coherence from the initial stages to subsequent modeling, flexibility becomes imperative in response to software-specific limitations. Despite the potential need for a format change, efforts are made to minimize such alterations to uphold the fidelity and consistency of the 3D model.

Ultimately, the Exportation Sub-Phase signifies the transition from the intricate processing realm to the subsequent modeling phase. This meticulous handling of data, formats, and standards encapsulates the commitment of the F.O.P.P.A. protocol to precision and reliability in archaeological documentation. It reinforces the protocol's foundational principles, emphasizing a methodical and standardized approach to ensure the scientific rigor and fidelity of the 3D models generated through the archaeological survey process.

Modeling

In the F.O.P.P.A. protocol, the Modeling Phase is the stage where the meticulously acquired data transforms into a coherent and visually accurate 3D model. Beginning with the Description Sub-Phase, the process involves attributing detailed characteristics to the model, defining the object's shape, surface features, and any distinctive elements identified during the survey. This initial step establishes a comprehensive and accurate representation of the archaeological subject. Following the Description Sub-Phase, the Elimination Sub-Phase becomes crucial in refining the model. Here, non-essential elements or artifacts inadvertently captured during the acquisition phase are strategically removed. This elimination process ensures that the model achieves a more focused and authentic representation, aligning with the intended archaeological interpretation. Moving forward, the Determination Sub-Phase involves a detailed analysis of the model's specific features. Parameters such as dimensions, proportions, and spatial relationships are precisely defined, ensuring accuracy in representing the archaeological subject. This step further aligns the digital model with the observed physical reality during the survey. The final step, the Distinction Sub-Phase, focuses on enhancing the visual clarity and interpretability of the 3D model. Specific elements, such as surface details, textures, or intricate features, are highlighted to facilitate a nuanced understanding of the archaeological subject. This careful distinction of key characteristics contributes to the model's fidelity and aids in conveying essential archaeological information. Throughout the Modeling Phase, adherence to established standards and best practices is paramount. The F.O.P.P.A. protocol emphasizes consistency in modeling techniques and the use of appropriate software tools. The phased approach, from Description to Distinction, ensures a systematic progression, minimizing the risk of oversights and inaccuracies.The Modeling Phase is not merely a technical transformation of data; it is a scientific interpretation translated into a digital realm. As such, it plays a crucial role in facilitating archaeological analysis, interpretation, and communication. The resulting 3D model becomes a tangible representation of cultural heritage, enabling researchers, scholars, and the public to engage with archaeological findings in a meaningful and immersive way.



Distinction & Determination

The "Distinction" sub-phase and the “Determination” sub-phase within the Modeling Phase hold paramount significance as they go beyond mere separation of components. It involves the meticulous identification and isolation of the specific object of the survey, acknowledging that not all detected elements may be pertinent to the project's objectives. This process discerns the essential components from external elements, such as those constituting the acquisition set or modern elements within a structural context, that are deemed irrelevant to preservation goals. The objective is not merely to segregate elements but to pinpoint and prioritize those integral to the project. This level of discrimination is crucial for refining the 3D model, ensuring that it encapsulates only the targeted components and aligns with the project's objectives. Thus, the "Distinction" sub-phase goes beyond technical separation; it is a thoughtful curation step, ensuring that the final 3D model encapsulates precisely what is essential for the project's research or preservation goals.

Description

The "Description" phase constitutes a pivotal step within the modeling process. Noteworthy is the deliberate omission of specific software recommendations, as these operational decisions are internal project choices. Contrarily, the modeling context prioritizes conceptual considerations. The "Description" phase emerges as a critical juncture wherein observations derived from the "Determination" and "Description" stages are meticulously documented. This step encapsulates the essence of the model, translating conceptual insights into a comprehensive representation. The emphasis on this phase underscores its significance in the meticulous translation of observational data into the conceptual framework of the 3D model.

In Database compile those properties: Modelling Note indicating 1)software used, 2)determination element to erase

Elimination

The "Elimination" sub-phase within the modeling process assumes a crucial role, strategically positioned at the conclusion of the modeling journey. Its intentional placement follows the "Description" phase, allowing for comprehensive and final considerations before undertaking the elimination. This deliberate sequencing underscores the significance of informed decision-making.

Elimination, in the context of 3D modeling, embodies the process of purposeful removal of elements that are extraneous to the survey or project objectives. This entails cleansing the 3D model of superfluous components, ensuring a refined and focused representation. The act of elimination demands precision, as it is an irreversible action with the potential to significantly impact the integrity of the model.

The consideration to place elimination as the concluding sub-phase is rooted in the principle of informed refinement. Following the meticulous description of the model in the preceding phases, the elimination step allows for a final review. This deliberate sequence permits a holistic assessment of the model's fidelity to the project's objectives and the survey's intrinsic elements. Consequently, any necessary adjustments or clarifications stemming from the description phase can be addressed before committing to irreversible elimination.

In essence, the "Elimination" sub-phase serves as a safeguard for the model's integrity. By systematically removing non-essential elements, it ensures that the final 3D representation aligns precisely with the intended project goals. This strategic approach not only enhances the accuracy and relevance of the model but also reinforces the need for a thoughtful and deliberate modeling process.

Export

The "Export" phase within the FOPPA protocol marks the culmination of the modeling journey, where the meticulously crafted 3D model transitions from the modeling software to its designated destination. This phase underscores the importance of strategic decision-making, requiring practitioners to align their choices with the ultimate purpose of the 3D model. Before initiating the export process, it is imperative to ascertain the final use of the 3D model. This consideration serves as a guiding principle, influencing the choice of format and ensuring compatibility with the intended application. For instance, if the model is destined for a database, practitioners must meticulously check and adhere to the formats supported by the database system. This proactive approach ensures seamless integration into the chosen database environment. Similarly, when the 3D model is intended for multimedia interfaces, practitioners must assess compatibility and recognize the spatial and RAM constraints of the intended platform. This foresight prevents potential issues related to file size, rendering efficiency, and overall performance. It becomes paramount to understand the technical specifications of the multimedia interface to optimize the user experience and prevent any unforeseen complications. In scenarios where the 3D model is destined for analytical purposes, collaboration with colleagues becomes a critical consideration. Verifying that the chosen export format is compatible with the tools and software employed by collaborators ensures a smooth exchange of data within the research or analysis context. Effective communication about the selected format becomes essential for cohesive collaboration and data interpretation. The export phase is not merely a technical step; it encapsulates a strategic decision-making process. It prompts practitioners to consider the broader implications of their choices, ensuring that the 3D model seamlessly integrates into its intended ecosystem. This holistic approach aligns with the overarching philosophy of FOPPA, emphasizing meticulous planning and precision throughout the entire survey and modeling workflow. In conclusion, the "Export" phase serves as a pivotal bridge between the intricacies of modeling software and the diverse landscapes of application. By considering the final use of the 3D model and aligning export choices accordingly, practitioners can enhance the utility, accessibility, and collaborative potential of their meticulously crafted models within the broader academic and practical contexts.

Archiving & Indexing

The "Archiving and Indexing" phase within the FOPPA Manual stands as the systematic and meticulous conclusion to the intricate journey of survey, modeling, and processing. This phase recognizes the significance of not only assigning a name to the 3D model but also ensuring a comprehensive organization of folders and subfolders. The organization mirrors the theoretical and methodological processes integral to the survey's creation. The method implemented in FOPPA, unlike traditional archaeological research, avoids confusing interpretative classifications that are susceptible to change with evolving archaeological interpretations. Instead, it introduces a classification structure grounded in the timing of discovery/initiation of the investigation and the morphology of the find. This approach guarantees straightforward recognition and verification of the data and metadata organization. Upon creating the name for the 3D model, the FOPPA method incorporates a systematic nomenclature. The name includes a letter of the transliterated Greek alphabet to denote the relative chronology, systematically aligning with the excavation campaigns. This adherence to the Greek alphabet provides an immediate chronological context to the model.

To ensure efficient organization and sharing of project data, a structured folder system is recommended, focusing on both the data production process and the archaeological finds themselves. Upon completion, all project participants are encouraged to download the folders from the Online Drive for offline access.

Here's how to create the folder:

Relative Time: Begin by indicating the relative time of the survey, divided into distinct phases corresponding to excavation campaigns or distinct acquisition periods. This helps organize data based on temporal context.

Technique Used: Specify the technique employed during the survey, using the following scheme:

fs: Freehand spiral

lf: Linear freehand

ad: Aerial with drone

vl: Free vertical (with drone or pole)

sb: Spiral with rotating base

el: Linear with easel

sm: Structured mixed

mu: Unstructured mixed

rm: Microscopic with rotating base

Type of Object: Assign a letter to identify the type of archaeological object surveyed:

A: Small object (≤ 1 cm)

B: Medium object (1 cm to 30 cm)

C: Large object (≥ 30 cm)

D: Open architectural structure

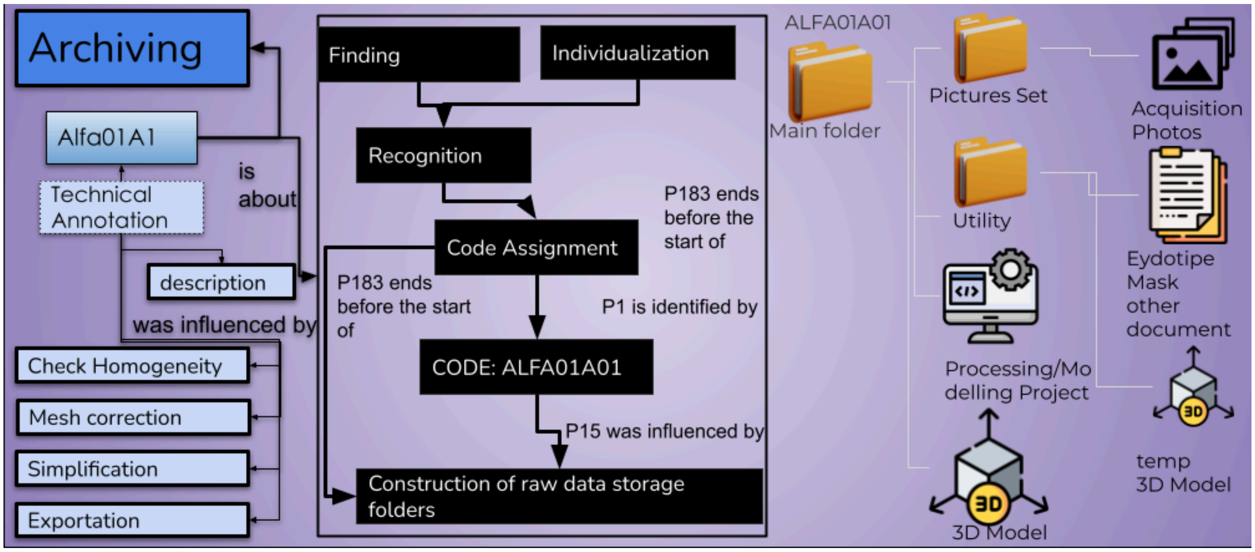
E: Architectural element

F: Cave or underground structure

G: Landscape element

Specific Finding Identifier: Allocate a unique number to each individual object surveyed. If a finding comprises multiple parts, each part should have its own identifier.

For example, a folder for the tenth finding, a 1 cm bead detected during the fourth phase of the project using a microscope, would be named DELTA09B10. This systematic approach ensures consistency and facilitates efficient retrieval of data during project analysis and dissemination.



Metadating: how to compile the database

When conducting a photogrammetric survey, it's crucial to gather comprehensive metadata to enrich the 3D model and facilitate future analysis. Here are 24 questions to consider during the survey process:

1. How is the acquisition set structured?

2. What is the average size of the monument surveyed?

3. How are the lights distributed in the acquisition set?

4. What type of camera was used?

5. What path was followed for the survey, and can it be drawn?

6. Was the process consistent for all finds?

7. Which processing software was used?

8. How many photos were collected per survey?

9. What matching software algorithm was used?

10. What algorithm/software was used for densification?

11. What algorithm/software was used to create the surface?

12. Was a single or multiple texture model generated?

13. What was the primary export format?

14. What was the average file size?

15. What was the average number of points per model?

16. Were simplification operations performed on the model?

17. Was subsequent modeling carried out, such as aligning point clouds?

18. Was there a homogeneous distribution of points in the point cloud?

19. Were non-inherent parts eliminated from the model?

20. Were there any points of non-coincidence between mesh and texture?

21. What was the final export format for the models?

22. How many measurements were taken and in what timeframe?

23. Where are the 3D models currently stored? Are they accessible online?

24. Was a specific storage method used?

These questions form the basis of the database card associated with each finding, containing properties such as archaeological name, internal classification, GPS coordinates, cultural reference, absolute datation, type of acquisition, processing software details, validation information, modeling notes, export notes, and media uploads of the Eidotype, 3D Mask, and 3D object. By meticulously documenting this information, researchers can ensure the integrity and accessibility of their photogrammetric data for future research endeavors.

Archaeological Name
A name given to the resource.
dcterms:title

Internal Classification *
An alternative name for the resource.
dcterms:alternative

Position UTM GPS *
The spatial or temporal topic of the resource, the spatial applicability of the resource, or the jurisdiction under which the resource is relevant.
dcterms:coverage

Cultural Reference
A related resource from which the described resource is derived.
dcterms:source

Absolute Datation
A point or period of time associated with an event in the lifecycle of the resource.
dcterms:date

Type of acquisition *
An established standard to which the described resource conforms.
dcterms:conformsTo

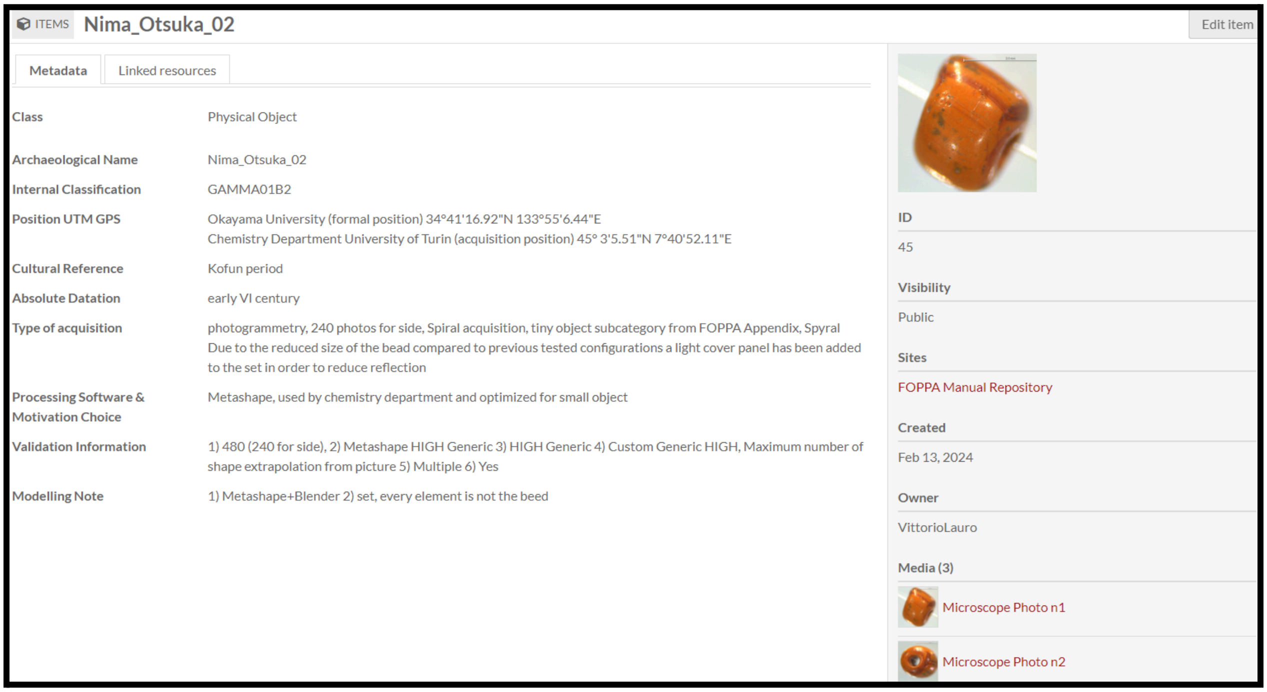
Processing Software & Motivation Choice *
The method by which items are added to a collection.
dcterms:accrualMethod

Validation Information *
1)"'n photos 2)algoritm/form used for match 3)algoritm/form used for densification 4) alg/for mesh reconstruction 5)texturing type 6) homogeneity check
dcterms:provenance

Modelling Note
1)software used 2)determination element to erase
dcterms:instructionalMethod

Export Note
1)Format and motivation 2)Alternative Storage Position
dcterms:tableOfContents

OMEKA-S card



ITEMS **Nima_Otsuka_02** Edit item

Metadata | Linked resources

Class	Physical Object
Archaeological Name	Nima_Otsuka_02
Internal Classification	GAMMA01B2
Position UTM GPS	Okayama University (formal position) 34°41'16.92"N 133°55'6.44"E Chemistry Department University of Turin (acquisition position) 45° 3'5.51"N 7°40'52.11"E
Cultural Reference	Kofun period
Absolute Datation	early VI century
Type of acquisition	photogrammetry, 240 photos for side, Spiral acquisition, tiny object subcategory from FOPPA Appendix, Spyral Due to the reduced size of the bead compared to previous tested configurations a light cover panel has been added to the set in order to reduce reflection
Processing Software & Motivation Choice	Metashape, used by chemistry department and optimized for small object
Validation Information	1) 480 (240 for side), 2) Metashape HIGH Generic 3) HIGH Generic 4) Custom Generic HIGH, Maximum number of shape extrapolation from picture 5) Multiple 6) Yes
Modelling Note	1) Metashape+Blender 2) set, every element is not the beed

ID
45

Visibility
Public

Sites
FOPPA Manual Repository

Created
Feb 13, 2024

Owner
VittorioLauro

Media (3)
Microscope Photo n1
Microscope Photo n2

OMEKA CLASSIC card

Archaeological Name

Nima_Otsuka_02

Internal Classification

GAMMA01B2

Position UTM GPS

Okayama University (formal position) 34°41'16.92"N 133°55'6.44"E
Chemistry Department University of Turin (acquisition position) 45° 3'5.51"N 7°40'52.11"E

Cultural Reference

Kofun period

Absolute Datation

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Processing Software & Motivation Choice

Metashape, used by chemistry department and optimized for small object

Validation Information

1) 480 (240 for side), 2) Metashape HIGH Generic 3) HIGH Generic 4) Custom Generic HIGH, Maximum number of shape extrapolation from picture 5) Multiple 6) Yes

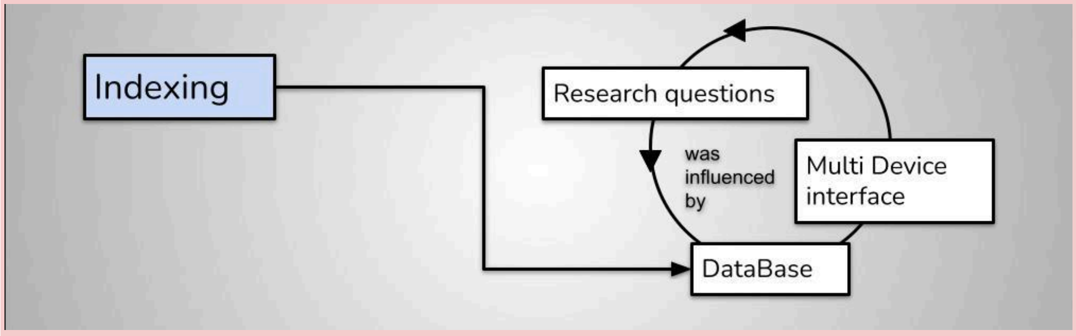
Export Note

1) Metashape+Blender 2) set, every element is not the beed

MEDIA

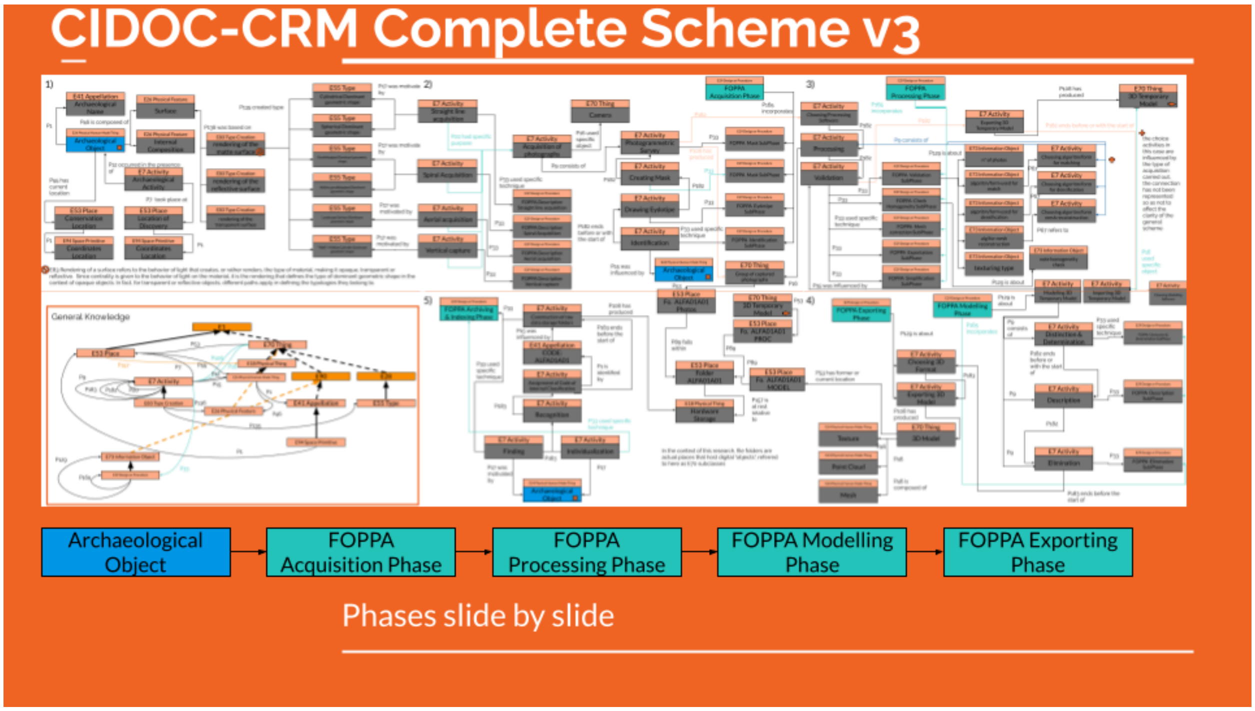
Research questions, Multidevice Interface, DataBase

Archiving and indexing, the final phase, meticulously organizes the 3D model and related files. The naming conventions provide a nuanced classification system, offering a chronological and morphological context that aligns with the excavation campaigns. The generative force behind this meticulous process is the research question. It's not merely a guiding principle; it's the very foundation upon which the entire project rests. The research question propels the acquisition phase, dictating what details are crucial for the overarching inquiry. Simultaneously, the findings from each phase feed back into the research question, generating new inquiries, perpetuating a cycle of discovery and exploration. As the 3D model crystallizes through the FOPPA protocol, the Multidevice Interface emerges as a crucial consideration. The model, born from meticulous steps, finds applications in diverse contexts. In educational settings, it might inhabit virtual museums accessible through browsers or fixed workstations, each requiring nuanced design considerations. In scientific contexts, the 3D model might serve in detailed archaeometric analyses or facilitate collaboration by allowing colleagues distant access. A key insight is the necessity of keeping the final interface in mind from the very beginning of the acquisition process. Choices made during acquisition reverberate through the interface, impacting accessibility, educational value, and scientific utility. The final piece of the puzzle is the Database. FOPPA distinguishes between the archive database, catering to scholars and those technically engaged, and the public database designed for broader accessibility. The CIDOC-CRM parameters that underpin the FOPPA protocol ensure adaptability to various database structures, facilitating seamless integration into scholarly and public domains.

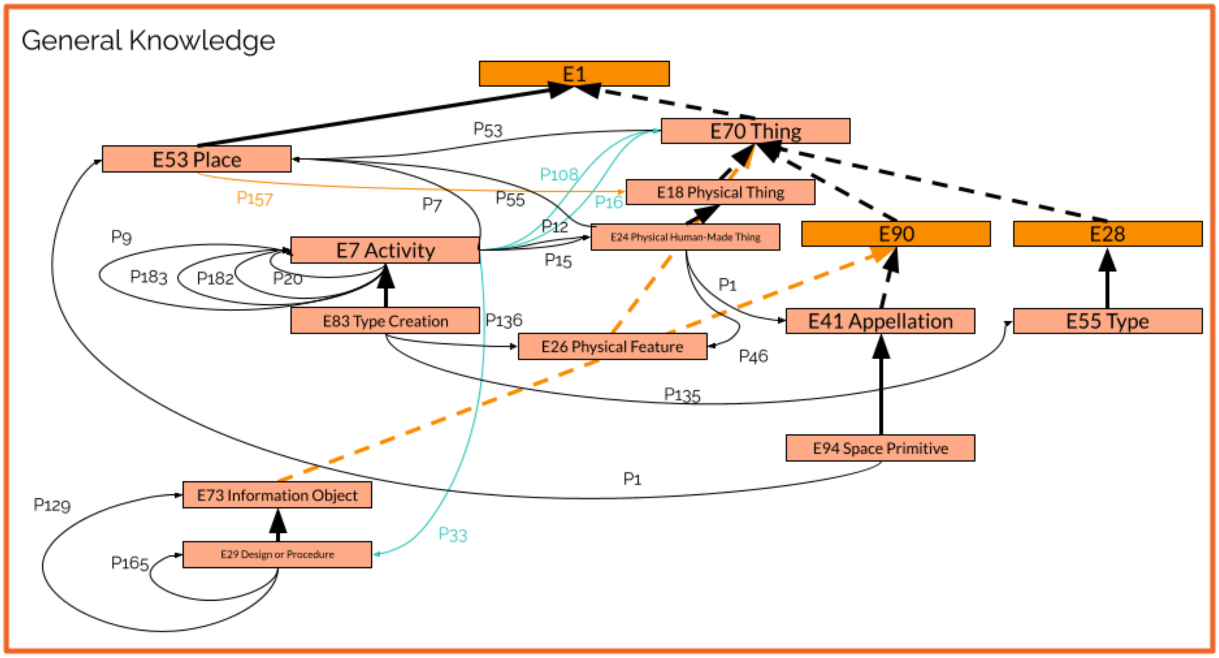


General Knowledge

In closing, we report the overall scheme of the FOPPA protocol, as well as the general knowledge that underlies the structure created in CIDOC-CRM and the total representative scheme as a verification tool for those who wish to adopt the FOPPA protocol even in contexts not described in this manual.



Complete FOPPA Protocol



General Knowledge FOPPA protocol in CIDOC-CRM of the highlighted examples there is an example three-dimensional model

type of object:

A. Small object (equal to or smaller than 1 cm)
B. Medium Object (1 cm to 30 cm)
C. Large Object (from 30 cm upwards)
D. Open architectural structure
E. Architectural element (part of an architectural structure)
F. Cave, rooms, or underground structure
G. Landscape element

A
Small objects mean those archaeological finds of any material that have a diameter equal to or less than 1 cm. Complete objects originally of small size fall into this category, as do fragments of small objects.
E.g.
the glass beads found in the Tobiotsuka Kofun of Okayama, Japan
the Minamikata bone nail, Okayama, Japan
individual fragments of the Löwenmensch figurines, Germany

B
Objects whose diameter is between 1 cm and 30 cm, even partial parts of a larger overall object. In the case of extremely irregular objects, consider the maximum extension diameter.
E.g.
Frame weight from Pyrgos Mavroraki, Cyprus
Boar jaw from Minamikata, Japan
an ushabti preserved in the Egyptian Museum, Turin

C
Objects whose diameter is equal to or greater than 30 cm. The objects can be removable or immovable, of any material but they must have the characteristic of not being able to access the inside, therefore they cannot be visited. The Statue of Liberty does not fall into this category.
E.g.
Jitsu Bosatzu of Mount Daizen, Japan
The monument to Ascanio Sobrero, Turin
The head of Nefertiti in Berlin, Germany

D
Architectural structures that can be visited or potentially visited, understood in their overall

external structure and not in their internal volumes. Courtyards and parade grounds fall into this category as they are external volumes.
E.g.
the Castle of Nola, Italy
the Tomb of the Treasure, Petra
the Colosseum, Rome

E
All those elements that are part of an architectural structure (D) both structurally (such as pillars and bastions) and as parts of an external structure (windows, portals and doors). Architectural elements may still be in place or may have been removed from the main structure or may never have been part of it.
E.g.
Column of the Roman Villa of Ossaia, Italy
Frieze of a temple from the sunken wreck of Hatsushima, Japan
The portal of the Cathedral of Orvieto, Italy

F
Caves, rooms and underground environments. Any finished practicable volume, including the environments that make up architectural structures (D).
E.g.
The quarries of Cornino Bay, Italy
the environments of Underground Naples, Italy
The meeting room on the third floor of Palazzo Nuovo in Turin, Italy

G
Landscape elements such as plains, woods, clearings, cliffs and any landscape element derived from the action of nature and anthropic action but which cannot be traversed within it. A rock structure does not fall into this category, but rather the facade is an architectural structure (D) and its interior is a walkable volume (F).
Dynamic or transforming landscape elements such as landslides, sinkholes, and sink holes also fall into this category. Elements of the landscape modified by human action but not visitable fall into this category. (So Mount Rashmure does NOT fall into this category, the Behistun carvings do fall into this category).
E.g.
The forest of Verbania, Italy
Sink hole in Casavatore, Italy
Montaguto landslide, Italy

technique used:

1. freehand spiral
2. linear freehand
3. aerial with drone
4. free vertical or with drone or with pole
5. spiral with rotating base
6. easel linear
7. structured mixed
8. mixed unstructured
9. rotating base microscope

Techniques 1-8 are described in the FOPPa manual in their specific paragraphs in the Acquisition Phase.

Technique 9 is described in Appendix 2 of the Foppa Manual

1. **Freehand Spiral:**
A systematic approach to capturing detailed information about individual finds and artifacts through spiral acquisition, allowing thorough documentation from various perspectives.

2. **Linear Freehand:**
An efficient method for surveying expansive structures by moving the camera along a straight line parallel to the structure, ensuring comprehensive coverage and minimal distortion.

3. **Aerial with Drone:**
The use of drones for capturing images from an elevated position, providing unique perspectives and facilitating efficient data collection for extensive territories or archaeological sites.

4. **Free Vertical or with Drone or with Pole:**
A method for capturing vertical images along tall structures, using drones or poles to ensure comprehensive documentation of structures with towering heights.

5. **Spiral with Rotating Base:**
A technique where the object rotates on a base while the camera remains stationary, simulating a spiral acquisition method. This approach is ideal for detailed documentation of small objects.

6. **Easel Linear:**

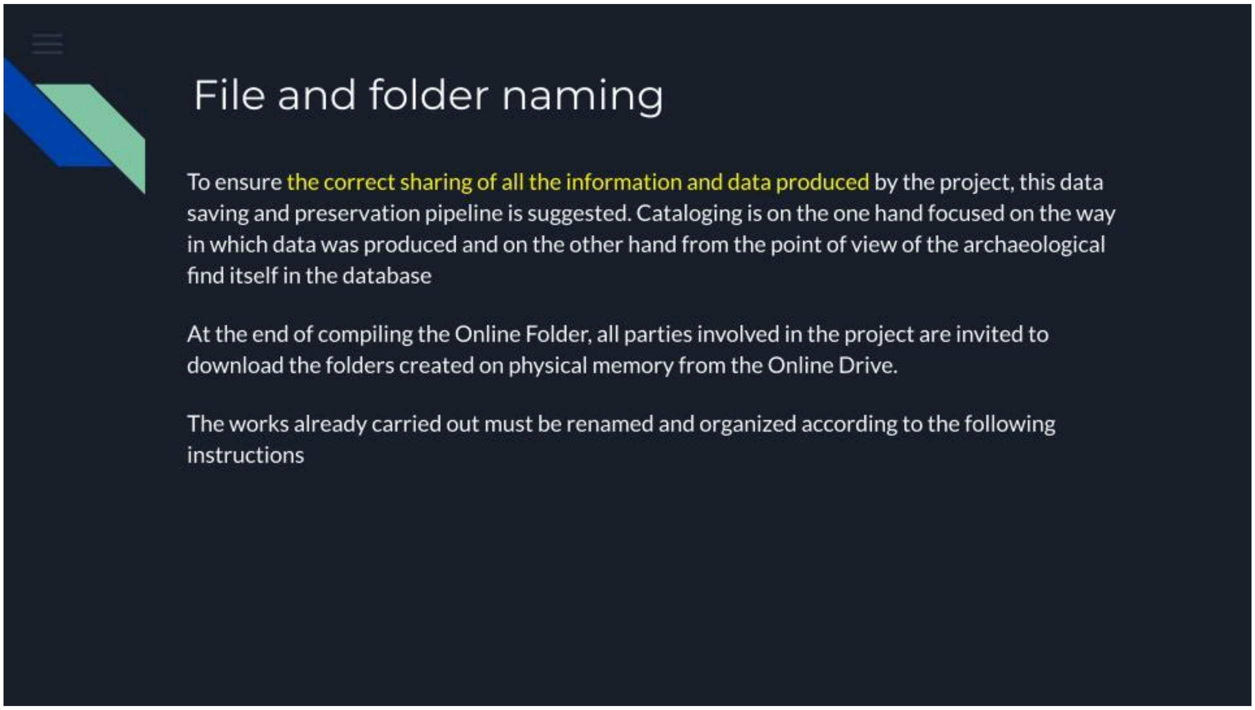
Acquisition with a camera fixed on a tripod, allowing systematic linear acquisitions with equidistant points. Suitable for wall analysis contexts where small but recognizable elements need to be documented.

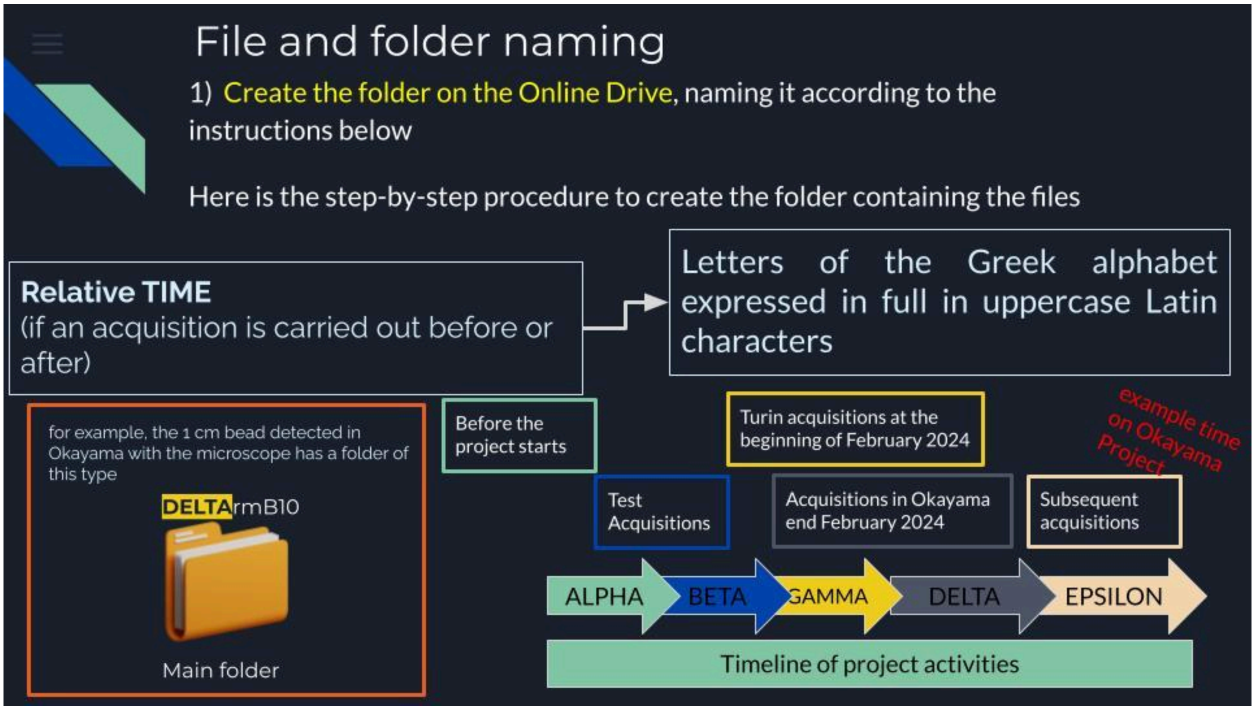
7. **Structured Mixed:**
A combination of spiral, linear, and vertical relief methods, ensuring comprehensive documentation while respecting the specific parameters of each acquisition module. Ideal for irregular small objects or body parts.

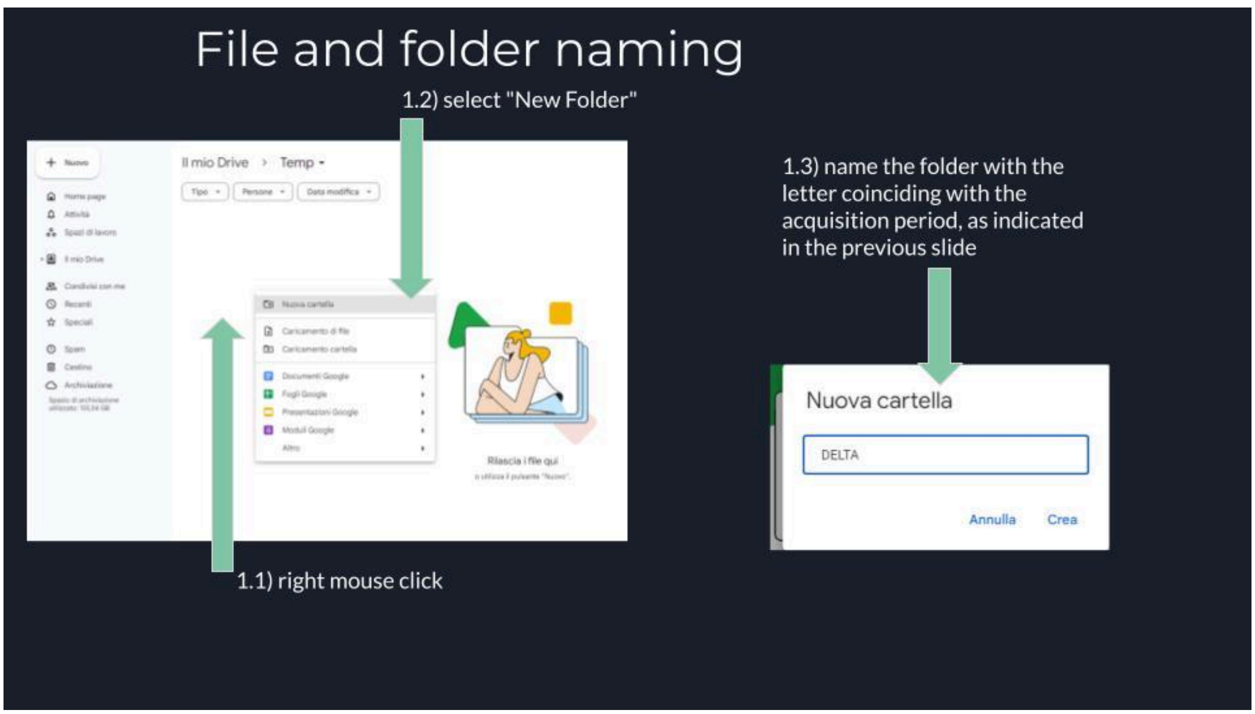
8. **Mixed Unstructured:**
An unstructured acquisition method for irregular environments or landscape elements where structured acquisition modules cannot be applied. Suitable for deep tunnels or narrow spaces where traditional methods are impractical.

9. **Rotating Base Microscope:**
Specific acquisitions for small or tiny objects using a microscope as a camera. Requires precise measures to overcome obstacles such as the orthographic lens of microscopes and rigorous management of acquisition points.

How to store data project according to the protocol




File and folder naming
1) Create the folder on the Online Drive, naming it according to the instructions below
Here is the step-by-step procedure to create the folder containing the files
Relative TIME
(if an acquisition is carried out before or after)
Letters of the Greek alphabet expressed in full in uppercase Latin characters
for example, the 1 cm bead detected in Okayama with the microscope has a folder of this type
DELTArmB10
Main folder
Before the project starts
Test Acquisitions
Turin acquisitions at the beginning of February 2024
Acquisitions in Okayama end February 2024
Subsequent acquisitions
example time on Okayama Project
ALPHA
BETA
GAMMA
DELTA
EPSILON
Timeline of project activities


File and folder naming
1.2) select "New Folder"
Il mio Drive
Temp
1.3) name the folder with the letter coinciding with the acquisition period, as indicated in the previous slide
Nuova cartella
DELTA
Annulla
Crea
1.1) right mouse click

File and folder naming

2) Now indicate the technique used during the survey

freehand spiral	fs
linear freehand	lf
aerial wifree	ad
vertical or with drone or with poleth drone	vl
spiral with rotating base	sb
easel linear	el
structured mixed	sm
mixed unstructured	mu
rotating base microscope	rm

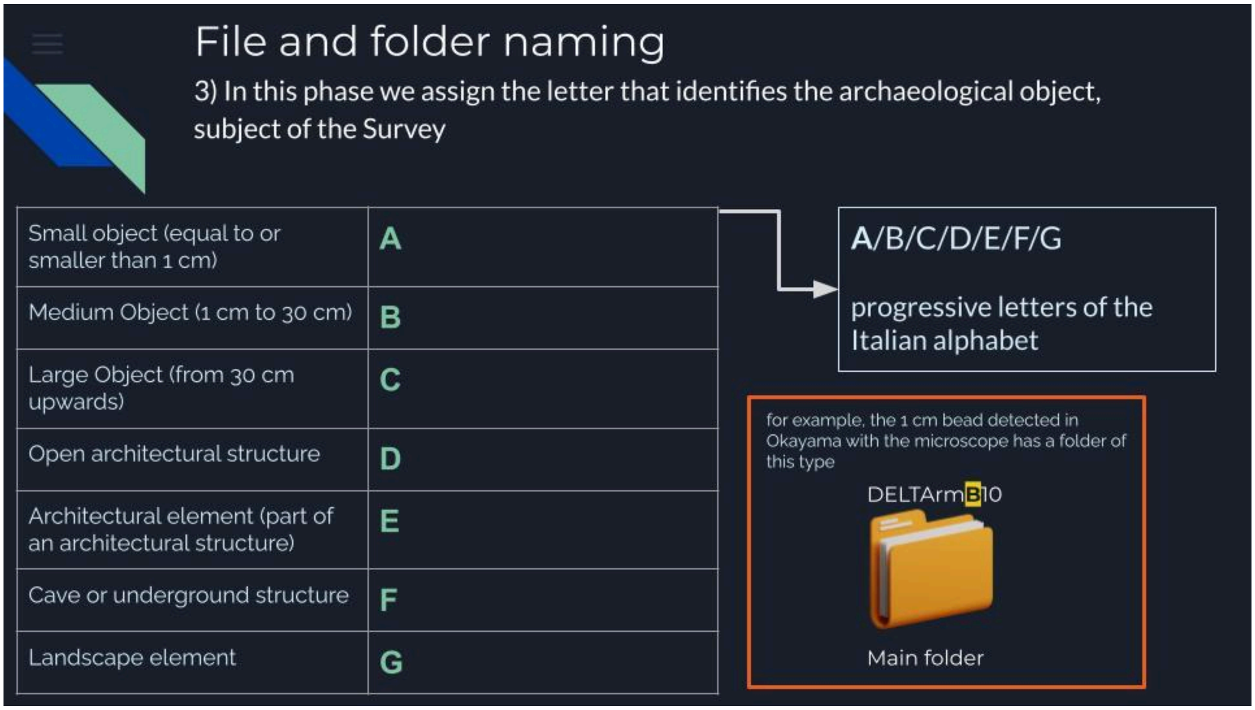
to be expressed with the reference code expressed in lowercase Latin letters

for example, the 1 cm bead detected in Okayama with the microscope has a folder of this type

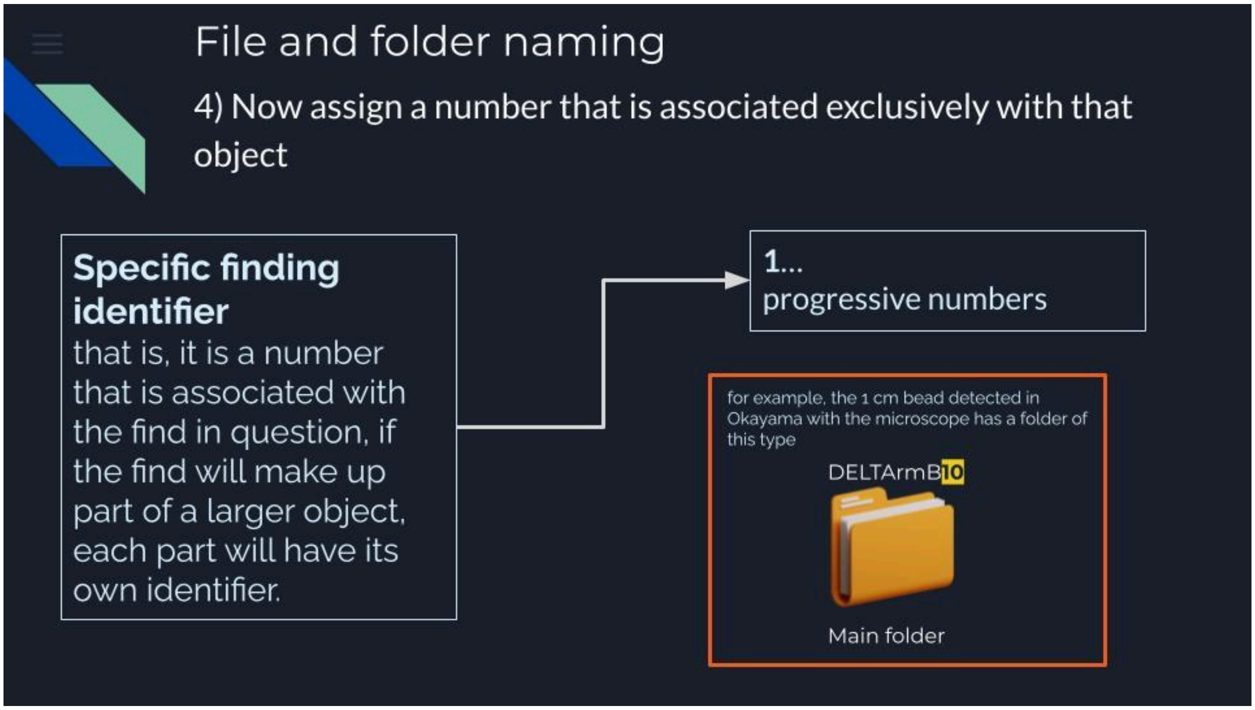
DELTArmB10

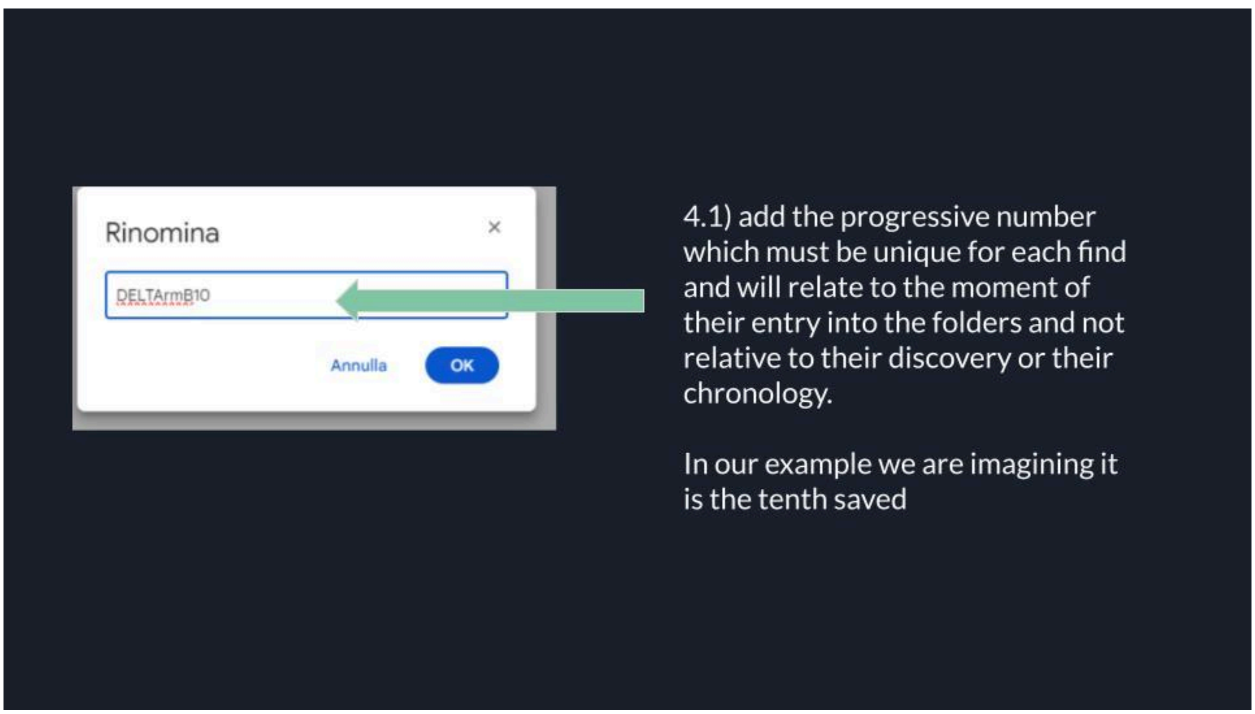
Main folder

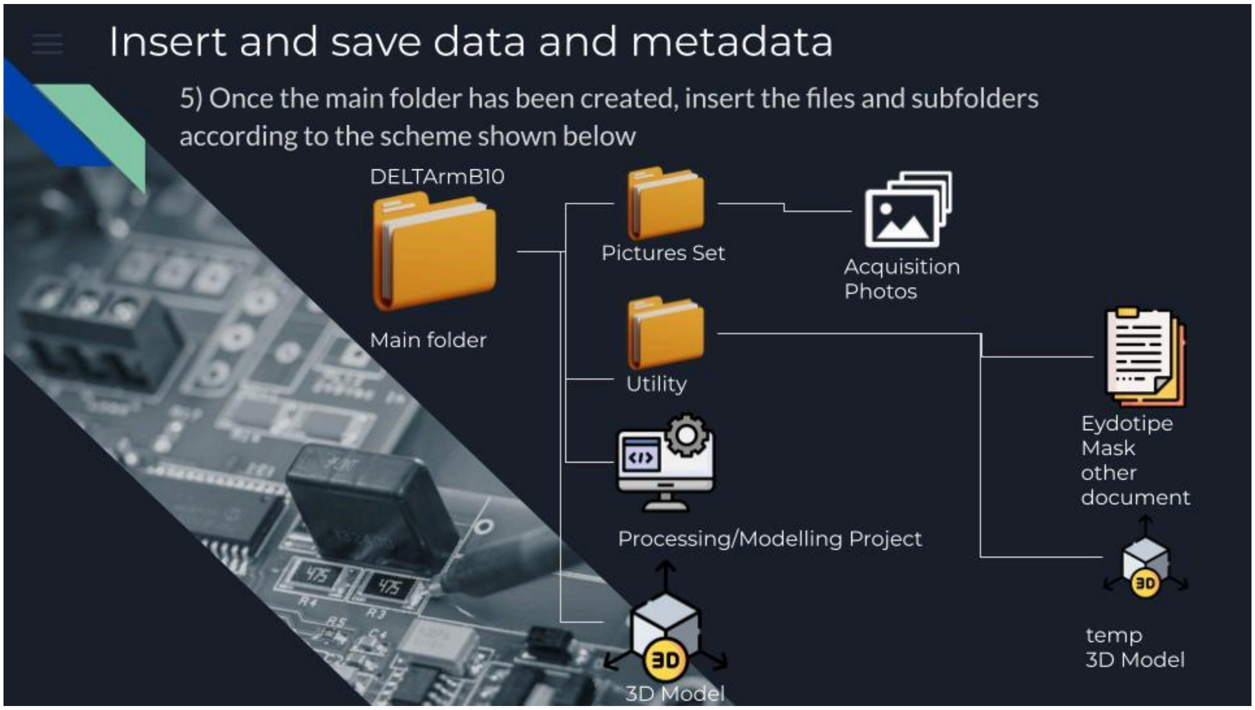
2.1) use the two-letter abbreviation coinciding with the code of the technique used during the acquisition

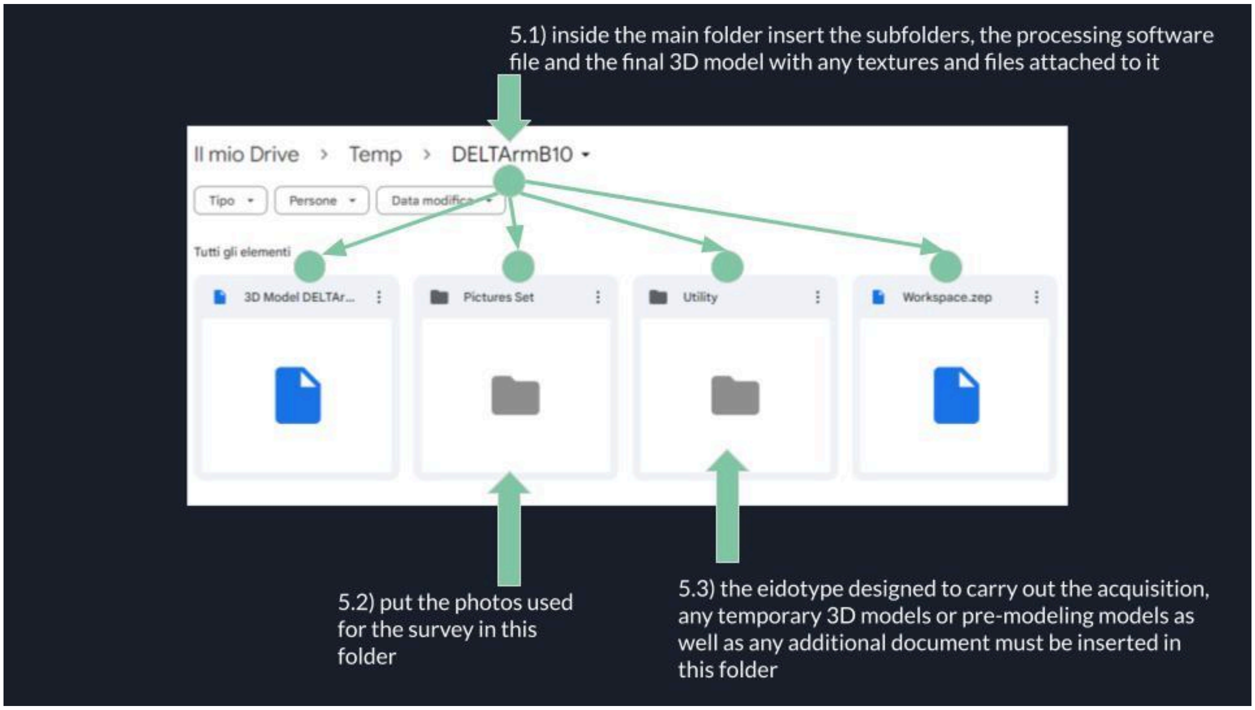

File and folder naming
3) In this phase we assign the letter that identifies the archaeological object, subject of the Survey
Small object (equal to or smaller than 1 cm)
A
Medium Object (1 cm to 30 cm)
B
Large Object (from 30 cm upwards)
C
Open architectural structure
D
Architectural element (part of an architectural structure)
E
Cave or underground structure
F
Landscape element
G
A/B/C/D/E/F/G
progressive letters of the Italian alphabet
for example, the 1 cm bead detected in Okayama with the microscope has a folder of this type
DELTArmB10
Main folder

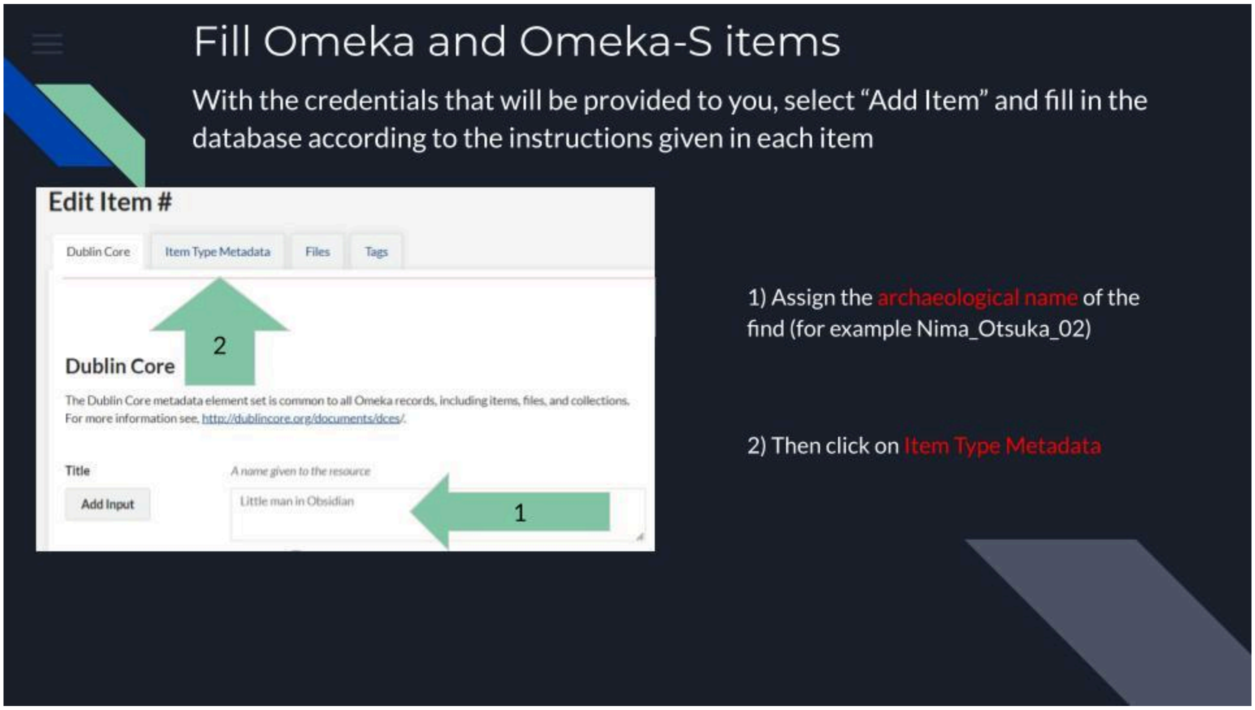

Rinomina
DELTArmB
Annulla
OK
3.1) add the letter relating to the type of object detected

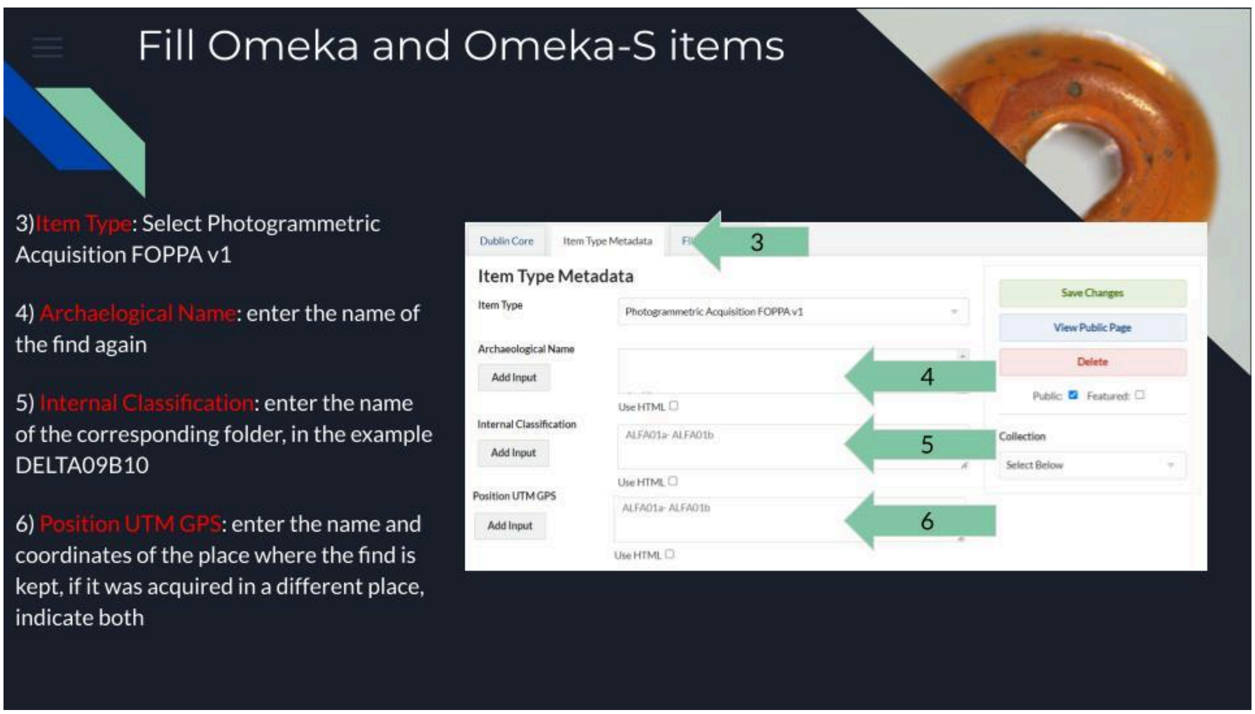

File and folder naming
4) Now assign a number that is associated exclusively with that object
Specific finding identifier
that is, it is a number that is associated with the find in question, if the find will make up part of a larger object, each part will have its own identifier.
1...
progressive numbers
for example, the 1 cm bead detected in Okayama with the microscope has a folder of this type
DELTArmB10
Main folder


Rinomina
DELTArmB10
Annulla
OK
4.1) add the progressive number which must be unique for each find and will relate to the moment of their entry into the folders and not relative to their discovery or their chronology.
In our example we are imagining it is the tenth saved


Insert and save data and metadata
5) Once the main folder has been created, insert the files and subfolders according to the scheme shown below
DELTArmB10
Main folder
Pictures Set
Acquisition Photos
Utility
Eydotipe Mask other document
Processing/Modelling Project
temp 3D Model
3D Model


5.1) inside the main folder insert the subfolders, the processing software file and the final 3D model with any textures and files attached to it
Il mio Drive › Temp › DELTArmB10
Tipo
Persone
Data modifica
Tutti gli elementi
3D Model DELTAr...
Pictures Set
Utility
Workspace.zep
5.2) put the photos used for the survey in this folder
5.3) the eidotype designed to carry out the acquisition, any temporary 3D models or pre-modeling models as well as any additional document must be inserted in this folder


Fill Omeka and Omeka-S items
With the credentials that will be provided to you, select "Add Item" and fill in the database according to the instructions given in each item
Edit Item #
Dublin Core
Item Type Metadata
Files
Tags
2
Dublin Core
The Dublin Core metadata element set is common to all Omeka records, including items, files, and collections. For more information see, http://dublincore.org/documents/dces/.
Title
A name given to the resource
Add Input
Little man in Obsidian
1
1) Assign the archaeological name of the find (for example Nima_Otsuka_02)
2) Then click on Item Type Metadata


Fill Omeka and Omeka-S items
3)Item Type: Select Photogrammetric Acquisition FOPPA v1
4) Archaeological Name: enter the name of the find again
5) Internal Classification: enter the name of the corresponding folder, in the example DELTA09B10
6) Position UTM GPS: enter the name and coordinates of the place where the find is kept, if it was acquired in a different place, indicate both
Dublin Core
Item Type Metadata
3
Item Type Metadata
Item Type
Photogrammetric Acquisition FOPPA v1
Archaeological Name
Add Input
4
Use HTML
Internal Classification
Add Input
ALFA01a- ALFA01b
5
Use HTML
Position UTM GPS
Add Input
ALFA01a- ALFA01b
6
Use HTML
Save Changes
View Public Page
Delete
Public:
Featured:
Collection
Select Below

Fill Omeka and Omeka-S items

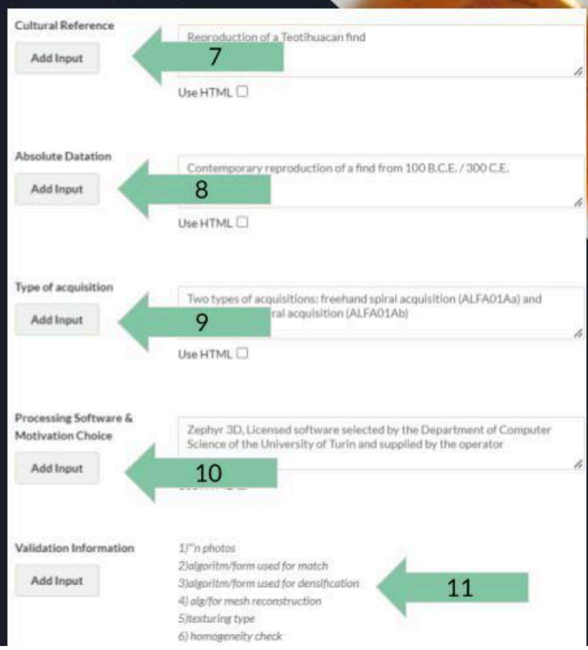
7) Cultural Reference: enter the cultural horizon of the find

8) Absolute Dating: the absolute dating of the find expressed in BCE years. and C.E.

9) Type of Acquisition: A detailed description of the acquisition mode

10)Processing Software & Motivation Choice: Which software was used and why that software

11) Validation Information: enter the information reported following the bulleted order shown in the form.



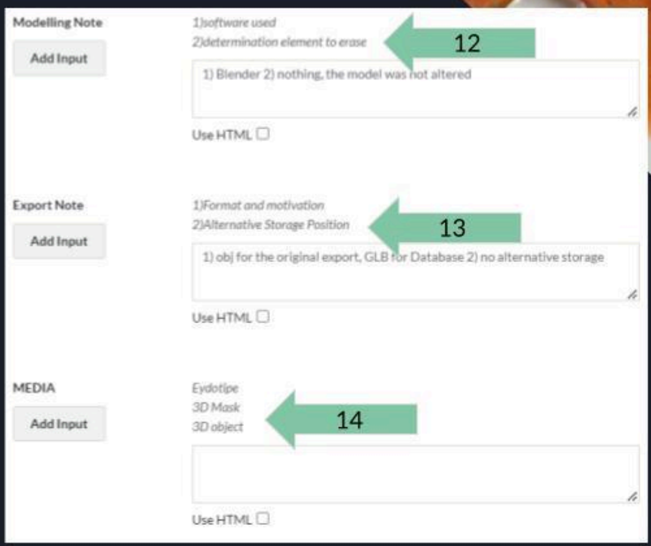
Fill Omeka and Omeka-S items

12) Modeling Note: fill in following the bulleted order shown on the sheet

13) Export Note: fill in following the bulleted order shown on the sheet

14) MEDIA: for space reasons do not upload the Media to this database as the server cannot support all the data of all the finds, this information will instead be moved to the Omeka-s database

SAVE AND PUBLISH THE SHEET



Fill Omeka and Omeka-S items

Whenever any information is compiled in the Excel sheet the information will be transmitted to the Omeka-S Database and the 3D files loaded into it. The Database will be accessible online upon request

The online databases, the survey manual and all the information relating to the methodology will then be available from a dedicated website

Update the checklist

At this point all that remains is to go to the Excel sheet that will be shared and in which there will be 5 columns

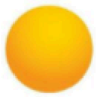
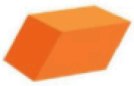
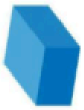
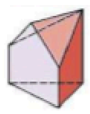
NAME OF EXHIBIT	CODE OF RECOGNITION	UPLOADED TO ONLINE DRIVE	COPIED TO PHYSICAL MEMORY	DATABASE TAB FILLED OUT
Enter the archaeological name of the find, for example Nima_Otsuka_02	Enter the identification code of the 3D Model for example BETA02B03	Enter the check when done	Enter the check when done	Enter the check when done

Thesaurus of Acquisition Types

freehand spiral	The Spiral Acquisition method in the F.O.P.P.A. protocol systematically captures detailed information by focusing on an object's center, starting with perpendicular shots, then 45° and 75° inclinations. It effectively documents finds from multiple perspectives, emphasizing intricate details and adapting to cavities or corridor-like structures. Further information in the FOPPA Manual
linear freehand	The Straight Line Acquisition method in the F.O.P.P.A. protocol surveys large buildings by moving the camera parallel to the structure while keeping it perpendicular. It ensures comprehensive coverage and captures intricate details. Maintaining at least 70% overlap between photos is crucial for accurate 3D model reconstruction and minimizing missed details. Further information in the FOPPA Manual
aerial	The Aerial Acquisition technique in the F.O.P.P.A. protocol involves moving parallel to the ground with a downward-tilted camera, ideal for drones. It effectively documents archaeological trenches and extensive territories, providing comprehensive, elevated images for detailed data collection and insights into spatial layout and features. Further information in the FOPPA Manual
vertical or with drone or with a pole	Vertical capture in the F.O.P.P.A. protocol involves systematic image acquisition along a vertical plane, ideal for tall structures. Drones or cameras on poles are recommended for flexibility and efficiency, ensuring comprehensive documentation of elevations like architectural elements and monuments. Further information in the FOPPA Manual
spiral with rotating base	The rotating base survey in the F.O.P.P.A. protocol involves rotating the object on a neutral-colored base with four camera positions (0°, 15°, 45°, 75°). The base rotates 5-15° between photos, ensuring detailed capture. Reflective objects require an absorbent background like velvet. Further information in the FOPPA Manual
easel linear	Acquisition with a camera fixed on a tripod involves moving the camera laterally at equidistant points, perpendicular to the object. It's useful for analyzing walls with small, colorimetrically distinct features. Each photo must overlap at least 70% with the preceding one, requiring numerous acquisitions. Further information in the FOPPA Manual
structured mixed	Structured mixed acquisition combines linear, spiral, and vertical surveys, adhering to the parameters of each method. While generally avoided for large structures, it's effective for irregular small objects and body parts. Specific guidelines for human body part surveys are detailed in Appendix #1 of the FOPPA Manual.
mixed unstructured	Unstructured mixed acquisition is used for irregular environments where standard modules don't apply. It requires photos to share at least 80% of representation, maintaining camera orientation. This method is necessary for areas like deep tunnels and involves taking numerous photographs to ensure detailed coverage. Further information in the FOPPA Manual
rotating base microscope	Rotating base acquisitions with a microscope camera are tailored for objects smaller than 2cm. Specific measures address challenges like orthographic lens effects and precise acquisition point management. Appendix #2 of the FOPPA Protocol extensively covers these methods for acquiring tiny objects.

Thesaurus of Dominant Geometric Shapes

By dominant geometry we mean the geometry assumed by the conjunction of the opposite vertices of the object according to an approximation based on the inclination of the faces not exceeding 45 degrees. This distinction is necessary for the application of the "rule of opposites" as defined in the FOPPA Manual. For example, a basketball is a sphere, a bottle is a cylinder, a book is a parallelepiped, a κρατήρ is a bowl/hemisphere. As an approximation, however, we can say that an irregular glass bead is a sphere, a ceramic shard or an animal jaw is a parallelepiped, a column is a cylinder and a crater or a lake are hemisphere.

	Sphere	A sphere is a perfectly round geometrical object in three-dimensional space, like a ball. Examples include a basketball or a glass bead.
	Parallelepiped	A parallelepiped is a six-faced figure (also called a polyhedron) where each face is a parallelogram. Examples include a book or a ceramic shard.
	Hemisphere	A hemisphere is half of a sphere, typically divided by a plane passing through its center. Examples include a crater or a bowl.
	Cylinder	A cylinder has two parallel circular bases connected by a curved surface. Examples include a column or a bottle.
	Plane Surface	A plane surface is a flat, two-dimensional surface extending infinitely in all directions. Examples include a flat stone tablet or a pavement segment.
	Irregular shape/Landscape	An irregular shape does not fit neatly into standard geometric categories and has uneven or asymmetrical surfaces. Examples include an irregular animal remains like Lyuba's mummy or a piece of pottery with an unusual form.

Thesaurus of Object Scales

Small object (equal to or smaller than 1 cm)	Small objects mean those archaeological finds of any material that have a diameter equal to or less than 1 cm. Complete objects originally of small size fall into this category, as do fragments of small objects.
Medium Object (1 cm to 30 cm)	Objects whose diameter is between 1 cm and 30 cm, even partial parts of a larger overall object. In the case of extremely irregular objects, consider the maximum extension diameter.
Large Object (from 30 cm upwards)	Objects whose diameter is equal to or greater than 30 cm. The objects can be removable or immovable, of any material but they must have the characteristic of not being able to access the inside, therefore they cannot be visited. The Statue of Liberty does not fall into this category.
Open architectural structure	Architectural structures that can be visited or potentially visited, understood in their overall external structure and not in their internal volumes. Courtyards and parade grounds fall into this category as they are external volumes.
Architectural element (part of an architectural structure)	All those elements that are part of an architectural structure (D) both structurally (such as pillars and bastions) and as parts of an external structure (windows, portals and doors). Architectural elements may still be in place or may have been removed from the main structure or may never have been part of it.
Cave, rooms, or underground structure	Caves, rooms and underground environments. Any finished practicable volume, including the environments that make up architectural structures
Landscape element	Landscape elements such as plains, woods, clearings, cliffs and any landscape element derived from the action of nature and anthropic action but which cannot be traversed within it. A rock structure does not fall into this category, but rather the facade is an architectural structure (D) and its interior is a walkable volume (F). Dynamic or transforming landscape elements such as landslides, sinkholes, and sink holes also fall into this category. Elements of the landscape modified by human action but not visitable fall into this category. (So Mount Rashmure does NOT fall into this category, the Behistun carvings do fall into this category).

Publications and cases study

Publications Regarding Methodology from an abstract point of view

1. V. Lauro, V. Lombardo, “Interdisciplinary approach to the dynamics of data collection, cataloging and homologation of photogrammetric models, in AIAr2022 Congresso Tematico_Padova, La sostenibiliá nei Beni Culturali, 29 June-1 July, 2022
2. Lauro, V.; Lombardo, V. The Cataloging and Conservation of Digital Survey in Archaeology: A Photogrammetry Protocol in the Context of Digital Data Curation. Heritage 2023, 6, 3113-3136. https://doi.org/10.3390/heritage6030166

Publications on the frontend and final query of the data produced according to the protocol

1. V. Lauro, V. Lombardo “CAVE, Archeology and virtual reality: visiting unreachable places” in AIAr2022 Congresso Tematico_Padova, La sostenibiliá nei Beni Culturali, 29 June-1 July, 2022
2. V. Lauro, V. Lombardo “Prototype of virtual reality and CAVE installation for transdisciplinary archeology” in 5th International Conference on Interaction Design and Digital Creation / Computing, IIAI AAI 2022, July 2-7, 2022, Kanazawa, Japan Honorable Mention; 10.1109/IIAIAAI55812.2022.00116
3. Lombardo, V., Lauro, V., Murtas, V., Goud, S. (2023). Merging Archaeological Site Recreation and Museum Exhibition. In: Holloway-Attaway, L., Murray, J.T. (eds) Interactive Storytelling. ICIDS 2023. Lecture Notes in Computer Science, vol 14384. Springer, Cham. https://doi.org/10.1007/978-3-031-47658-7_6

Publications on a case study

In relation to a project acquired according to the protocol

1. Lauro Vittorio, Lombardo Vincenzo, A Digital Data Curation-Based Photogrammetric Acquisition Methodology for Cultural Heritage, expanded with CIDOC CRM Compatibility: protocol BeA-PG, GCH 2023 - Eurographics Workshop on Graphics and Cultural Heritage, 2023, https://doi.org/10.2312/gch.20231169

Regarding the recovery of closed or obsolete projects

1. R.A.O. Project Recovery: Methods and Approaches for the Recovery of a Photographic Archive for the Creation of a Photogrammetric Survey of a Site Unreachable over Time doi: 10.3390/heritage6060250